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In this chapter, we will learn about

- the structure of an atom
- three types of radioactive decay
- decay curves
- applications of radionuclides

Atomic Structure and Radioactive Decay

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2.1 The atomic model

1 Structure of an atom

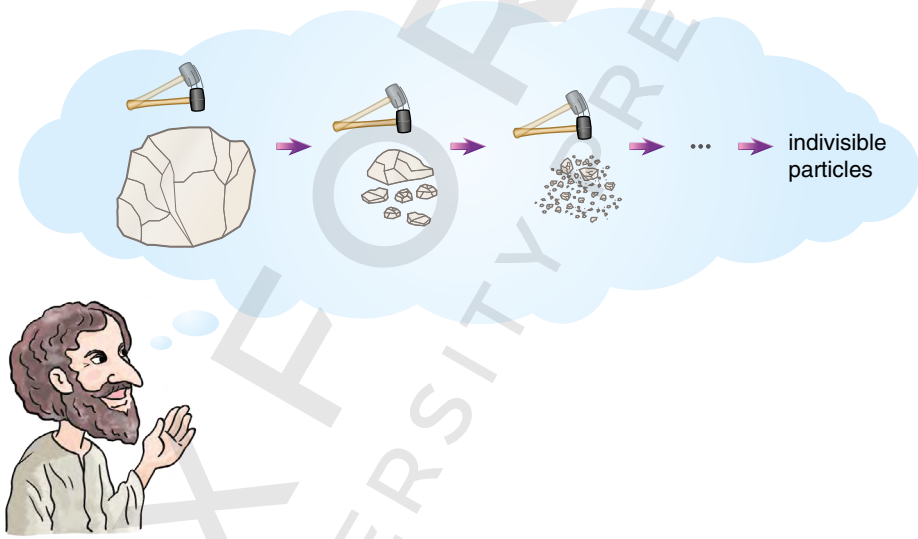
Flipped classroom
Watch this and answer the questions.

DSE goal
Describe the structure of an atom.

Pre-study **Atomic model**

We learned about the atomic model in junior science. Let us revise this concept.

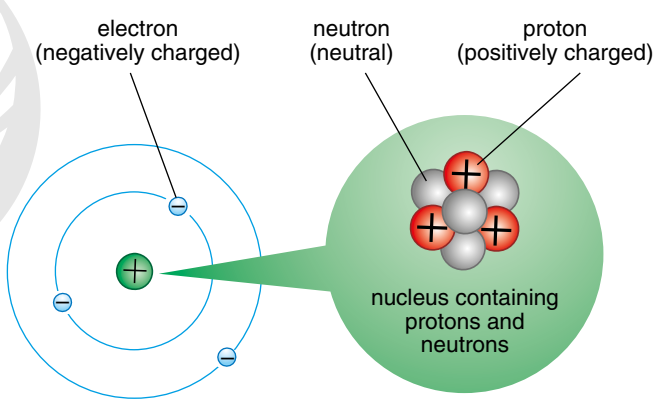
The idea of the atom has been developed for more than 2000 years. Ancient Greeks first proposed that matter was made up of indivisible particles—atoms.



Ancient Greeks proposed the idea of atom (an indivisible particle).

With the advancement of technology, scientists found that the particle they called an atom actually consists of **protons**, **neutrons** and **electrons**. Protons and electrons are positively and negatively charged respectively, while neutrons are neutral. A neutral atom contains the same number of protons and electrons.

The figure below shows a simplified atomic model. In the model, an atom is mostly empty space. There is a tiny core at the centre of an atom called a nucleus. It contains all the protons and neutrons of the atom and is thus positively charged. The electrons of the atom orbit around the nucleus.



A simplified atomic model.

proton 質子　neutron 中子　electron 電子

✎ The existence of neutron was first postulated by Rutherford. It was finally discovered in 1932 by Sir James Chadwick (1891–1974). He won the Nobel Prize for Physics in 1935.

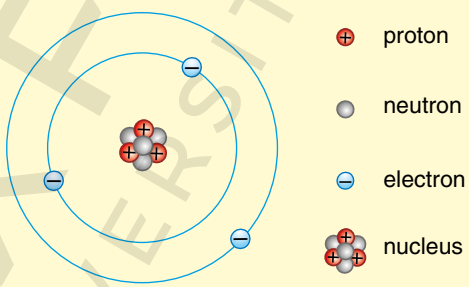
Protons, neutrons and electrons make up an atom and are called *subatomic particles*. Protons and neutrons are collectively called **nucleons**. They are nearly of the same mass. The mass of an electron is negligible when compared to that of a nucleon (Table 2.1a). This explains why the nucleus takes up most of the mass of an atom.

Subatomic particles	Position in atom	Approximate size	Relative charge	Relative mass
proton	inside nucleus	10^{-15} m	1	1
neutron	inside nucleus	10^{-15} m	0	1
electron	orbiting around nucleus	$< 10^{-18}$ m	–1	1/1800

Table 2.1a *Comparing a proton, a neutron and an electron.*

✎ The relative sizes of electron, proton/neutron, nucleus and atom are:
electron : proton : nucleus : atom
≈ 1 mm : 1 m : 10 m : 100 km
Guide Ss to think that most of the region in an atom is empty space.

In an atom, the nucleus consists of protons and neutrons with electrons orbiting around it. The rest of the atom is largely empty space.

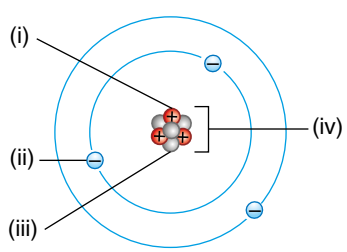


Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 The figure below shows an atomic model.



(a) Name structures (i) to (iv).

(b) State the electric charges of (i) to (iii).

2 Protons and neutrons are called _____.

3 *True or false:* There is a lot of empty space in an atom. (T/F)

4 *True or false:* Electrons make up about half of the mass of an atom. (T/F)

subatomic particle 亞原子粒子 nucleon 核子

Nature of science

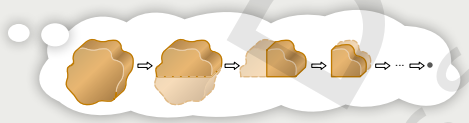
Scientific model

You learned that an atom is made of a positively charged nucleus and the surrounding electrons. Where did this idea come from?

Around 500 BC



Democritus



I believe we cannot divide matter into smaller pieces endlessly. We would eventually get to the smallest piece of matter called an atom.

Early 1800s

I agree with you. All matter is composed of indivisible atoms. But, from the results of our experiments, there should be different types of atoms.



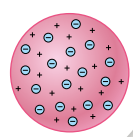
John Dalton

1897



J. J. Thomson

Well, an atom of any type should contain a cloud of positive charge, in which small negatively charged electrons are embedded.



1911

You are wrong! An atom should have a tiny positive core with the electrons orbiting around it.





Ernest Rutherford

Scientists often propose scientific models to explain observations or experimental results. When a model fails to explain new observations, for example, obtained due to more advanced and precise instruments, scientists will revise old models or develop new models that are consistent with the new observations.

Normally the scientific model that gives the best prediction of observations will be accepted by the scientific community as the standard model.

Democritus 德謨克利特　John Dalton 道爾頓　J. J. Thomson 湯姆生

DSE goal

Define atomic number and mass number.

Use symbolic notations to represent nuclides.

2 Atomic number and mass number

In the rest of this book, we will frequently use two terms, the **atomic number** (Z) and the **mass number** (A), to represent the composition of an atom.

The atomic number (Z) is the number of protons in the nucleus of an atom. The mass number (A) is the total number of protons and neutrons in the nucleus of an atom.

Note that ($A - Z$), i.e. the difference between A and Z, represents the number of neutrons in the nucleus. ▶

Atoms of the same *element* have the same number of protons. In other words, each element has a unique atomic number. However, atoms of the same element can have different numbers of neutrons, and hence different mass numbers. Table 2.1b shows some elements and their atomic numbers.

Element (Symbol)	Z
Hydrogen (H)	1
Helium (He)	2
Carbon (C)	6
Lead (Pb)	82
Radon (Rn)	86
Radium (Ra)	88
Thorium (Th)	90
Uranium (U)	92





Table 2.1b *Some elements and their atomic numbers.*

Simulation

→ This simulation shows graphically the Rutherford's atomic model for some elements. On choosing a certain atomic number, the graphic model and some relevant information of the element will be displayed.

We can represent an atom of an element with the notation ${}^{A}_{Z}X$. In the notation, X is the chemical symbol of the element; Z and A are the atomic number and mass number of the atom respectively. Table 2.1c shows the structures and notations of some atoms.

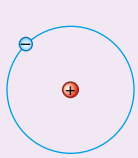
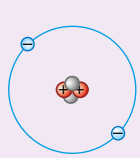
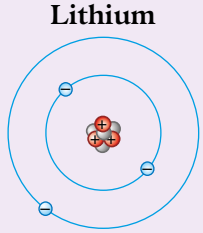
	Hydrogen	Helium	Lithium
			
Atomic number Z	1	2	3
Mass number A	1	4	7
Notation ${}^{A}_{Z}X$	${}^{1}_{1}\mathrm{H}$	${}^{4}_{2}\mathrm{He}$	${}^{7}_{3}\mathrm{Li}$
Number of neutrons ($= A - Z$)	0	2	4

Table 2.1c *The structures and notations of some atoms.*

atomic number 原子序數　　mass number 質量數　　element 元素

DSE goal

Define isotope.

DSE exam

12(1A)Q36

Recall that atoms of the same element have the same atomic number. The chemical symbol 'C' already hints at what the atomic number is.

The number of outermost electrons in an atom determines the chemical properties of a nuclide. Therefore, isotopes of an element have the same chemical properties.

On the other hand, the physical properties of a nuclide depend on the composition of the nucleus. Isotopes of an element may not have the same physical properties.

3 Nuclides and isotopes

Nuclide is a type of atoms with specific numbers of protons and neutrons. It is usually represented by its element name and mass number. For example, a carbon atom with mass number 12 can be represented by carbon-12 or C-12.

As mentioned before, atoms of an element may have different numbers of neutrons and hence different mass numbers. These are called **isotopes**. For example, H-1, H-2 and H-3 are three isotopes of hydrogen (Table 2.1d).

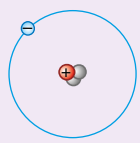
	Hydrogen-1	Hydrogen-2	Hydrogen-3
			
Atomic number Z	1	1	1
Mass number A	1	2	3
Notation ${}^{A}_{Z}X$	${}^{1}_{1}\mathrm{H}$	${}^{2}_{1}\mathrm{H}$	${}^{3}_{1}\mathrm{H}$
Number of neutrons ($= A - Z$)	0	1	2

Table 2.1d *Examples of isotopes.*

Isotopes are different atoms that contain the same number of protons but different numbers of neutrons.

Example 1 Identifying isotopes

Three different nuclides are shown below.

${}^{40}_{20}P$ ${}^{46}_{20}Q$ ${}^{46}_{22}R$

(a) Find the numbers of protons and neutrons in Q.

(b) Which two nuclides are isotopes of the same element? Explain briefly.

Solution

(a) Number of protons = Z = 20

Number of neutrons = $A - Z$ = 46 − 20 = 26

(b) P and Q have the same atomic number (20) but different mass numbers (40 and 46). Therefore, they are isotopes of the same element.

nuclide 核素　　isotope 同位素

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Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 The number of _____ in the nucleus of an atom is called its **atomic number**. It is represented by the letter _____.

2 The total number of _____ and _____ in the nucleus of an atom is called its **mass number**. It is represented by the letter _____.

3 Complete the following table.

	Element	Atomic number	Mass number	No. of protons	No. of neutrons	Notation
(a)	C	6	13			
(b)	Th					$^{238}_{90}\text{Th}$
(c)	Pb	82	208			
(d)	Pu					$^{244}_{94}\text{Pu}$
(e)	I			53	74	

4 Isotopes have the same number of (protons / neutrons) but different numbers of (protons / neutrons).

5 Three different nuclides are shown below.

$^{18}_{8}P$ $^{18}_{9}Q$ $^{16}_{8}R$

(a) Which nuclides are isotopes of the same element?

(b) Which nuclides have the same number of nucleons?

Exercise 2.1

1 A neutral atom has 3 electrons and 4 neutrons. How many protons does it have?

A 1 B 3

C 4 D 7

2 Which of the following statements about an atom are correct?

(1) It is composed of small particles.

(2) Most of its mass is concentrated in the nucleus.

(3) Its nucleus is electrically charged.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

3 Which kind of particle (proton, neutron or electron) in an atom

(a) has a positive charge?

(b) has the smallest mass?

(c) is uncharged?

4 Use the notation $^{A}_{Z}X$ to represent a calcium (Ca) nucleus with 20 protons and 23 neutrons.

★**5** A thorium nuclide is represented by $^{238}_{90}\text{Th}$.

(a) How many (i) protons, (ii) neutrons and (iii) electrons does a neutral $^{238}_{90}\text{Th}$ atom contain?

(b) What is the notation of another thorium nuclide that contains 2 more neutrons?

★**6** The following shows three neutral atoms.

P: 6 electrons, mass number 12

Q: 6 protons, 8 neutrons

R: 6 neutrons, atomic number 8

Which of them are isotopes of the same element? Explain briefly.

2.2 Radioactive decay

In Chapter 1, we learned about the nature of the three kinds of nuclear radiations, α, β and γ. All of them are emitted spontaneously by unstable nuclei. In this part, we will further study these three kinds of nuclear radiations by referring to the atomic structure.

1 Equation representation of decay

DSE goal

Realise the occurrence of radioactive decay in unstable nuclides.

Represent radioactive transmutations in α, β and γ decays using equations.

DSE exam

13(1A)Q34, 14(1A)Q31, 16(1B)Q9(a), 18(1B)Q10(a)(i), 20(1A)Q31, 21(1A)Q31, 21(1B)Q9(a), 23(1B)Q9(a)

The nuclides that can emit nuclear radiation are called **radionuclides**. They have too much energy and are unstable. They can release the excess energy **spontaneously** by emitting nuclear radiation. This kind of nuclear reaction is called **radioactive decay** or disintegration (Fig 2.2a).

It is possible that some isotopes of the same element are radioactive and some are not. For example, carbon-14 is radioactive while carbon-12 is stable. Isotopes that are radioactive are called *radioisotopes*.

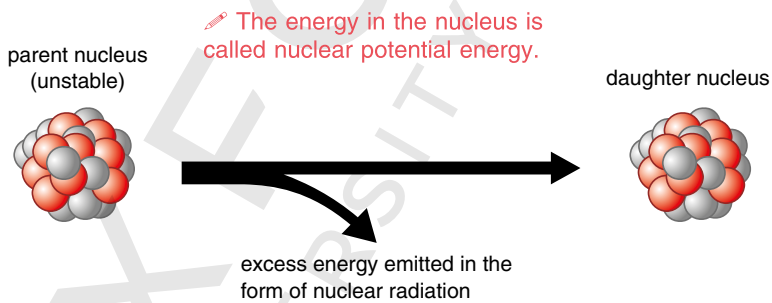


Fig 2.2a *Radioactive decay.*

Based on the types of nuclear radiation emitted, radioactive decay can be categorised into **alpha decay** (α decay), **beta decay** (β decay) and **gamma emission** (γ emission). In an α decay, a nucleus emits an α particle. In a β decay, a nucleus emits a β particle. In γ emission, a nucleus emits γ ray (Table 2.2a).

Gamma emission is also referred to as gamma decay.

	α particle	β particle	γ ray
Nature	helium nucleus (with 2 protons and 2 neutrons)	electron	electromagnetic wave
Notation ${}^{A}_{Z}X$	${}^{4}_{2}\mathrm{He}$	${}^{0}_{-1}\mathrm{e}$	(N/A)

An electron has a negligible mass compared with a proton or a neutron. Its mass number is 0.

Table 2.2a *Different types of nuclear radiation and their notations.*

radionuclide 放射性核素　radioactive decay 放射衰變　radioisotope 放射性同位素
alpha decay (α decay) α 衰變　beta decay (β decay) β 衰變　gamma emission (γ emission) γ 發射

Simulation

→ This simulation shows graphically α, β and γ decays. It is available on OUP web in PC version (.exe) only.

α and β decays are examples of **transmutation** in which the composition of a nucleus changes. For example, when uranium-238 emits an α particle ($^4_2$He), its mass number decreases by 4 and its atomic number decreases by 2. The nuclide changes into thorium-234 (Fig 2.2b).

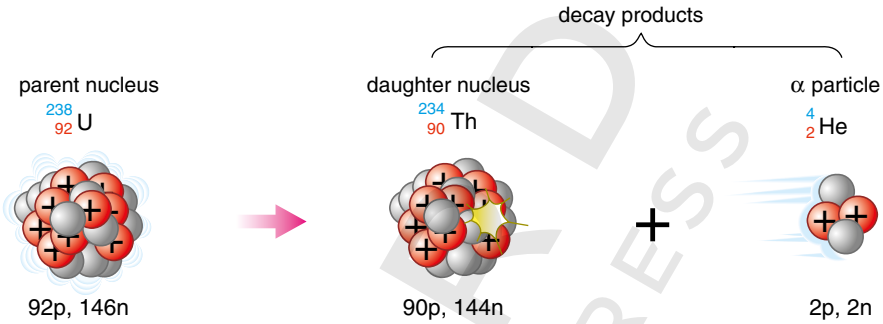


Fig 2.2b *An α decay of uranium-238.*

We may use an equation to represent a decay and indicate the composition of each nucleus and particle involved. The equation for the above decay is:

$$^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + {}^{4}_{2}\text{He}$$

Figure 2.2c shows an example of β decay. When thorium-234 emits a β particle ($^0_{-1}$e), its mass number stays the same but the atomic number increases by 1. The nuclide changes into protactinium-234.

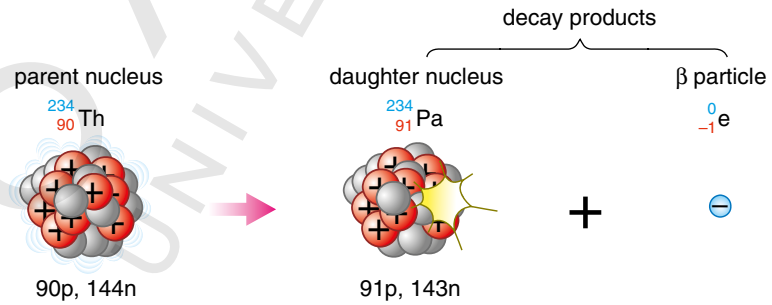


Fig 2.2c *A β decay of thorium-234.*

The equation for the decay is:

$$^{234}_{90}\text{Th} \rightarrow {}^{234}_{91}\text{Pa} + {}^{0}_{-1}\text{e}$$

$^1_0$n → $^1_1$p + $^0_{-1}$e

▶ Note that in a β decay, one of the neutrons in the parent nucleus becomes a proton and an electron. This electron is emitted as a β particle. Therefore, **a β particle comes from inside the nucleus**; it is not an electron orbiting around the nucleus.

transmutation 嬗變

✎ After emitting an α or a β particle, the daughter nucleus may still be in an excited state. The extra energy may be emitted as gamma ray.

In γ emission, a nucleus only emits γ ray and does not have any changes in its composition. Figure 2.2d shows an example of γ emission. Gamma emission is **not** a transmutation as there is no change in the composition of the nuclide.

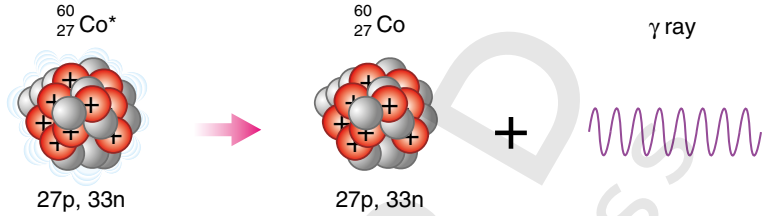


Fig 2.2d *A γ emission of cobalt-60.*

The equation for the decay is (* means the nucleus has more energy than normal):

The mass number and atomic number of the nuclide are unchanged in γ emission. ▶

$$^{60}_{27}\text{Co}^* \rightarrow {}^{60}_{27}\text{Co} + \gamma$$

Table 2.2b shows the equations for different decays undergone by the nucleus of $^{A}_{Z}X$.

	α decay	β decay	γ decay
Mass number of daughter nucleus	$A - 4$	A	A
Atomic number of daughter nucleus	$Z - 2$	$Z + 1$	Z
Equation representation ▶	$^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\text{He}$	$^{A}_{Z}X \rightarrow {}^{A}_{Z+1}W + {}^{0}_{-1}\text{e}$	$^{A}_{Z}X^* \rightarrow {}^{A}_{Z}X + \gamma$

The mass number and atomic number of the parent nucleus are equal to the sum of the mass numbers and the sum of the atomic numbers of the decay products respectively.

Table 2.2b *Equations for different decays.*

Example 2 An α decay of radium-226

Radium-226 ($^{226}_{88}\text{Ra}$) decays into radon (Rn) by emitting an α particle. Write down an equation for the decay process.

Solution

The α decay can be written as:

$$^{226}_{88}\text{Ra} \rightarrow {}^{A}_{Z}\text{Rn} + {}^{4}_{2}\text{He}$$

Consider the mass numbers and atomic numbers before and after the decay.

$$226 = A + 4 \qquad \text{and} \qquad 88 = Z + 2$$
$$A = 222 \qquad\qquad Z = 86$$

Therefore, the equation is

$$^{226}_{88}\text{Ra} \rightarrow {}^{222}_{86}\text{Rn} + {}^{4}_{2}\text{He}$$

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* The electron emitted in a β decay comes from the orbiting electrons of the radionuclide. (T/F)

2 When a radionuclide undergoes a β decay, its number of protons (remains unchanged / increases by 1 / decreases by 1) and its number of neutrons (remains unchanged / increases by 1 / decreases by 1).

3 Complete the following table.

	Parent nucleus (X)	Daughter nucleus (Y)	Type of decay
(a)	$^{226}_{88}X$		α
(b)	$^{214}_{83}X$		β
(c)	$^{60}_{28}X$		γ
(d)		$^{210}_{83}Y$	β
(e)		$^{230}_{90}Y$	α

Learn by practice 1

1 Write the equations for the following radioactive decay processes.

(a) A $^{230}_{90}$Th nucleus undergoes an α decay to form nucleus W.

(b) A $^{234}_{90}$Th nucleus undergoes a β decay to form nucleus X.

(c) Nucleus Y undergoes an α decay to form a $^{206}_{82}$Pb nucleus.

(d) Nucleus Z undergoes a β decay to form a $^{228}_{89}$Ac nucleus.

2 Decay series

As discussed before, a uranium-238 nuclide undergoes an α decay and changes into a thorium-234 nuclide. The daughter nuclide is also radioactive and in turn undergoes β decay to form a protactinium-234 nuclide. The process carries on until the stable lead-206 nuclide is formed. These sequential decays form a *decay series*.

The above decay series can also be represented by equations:

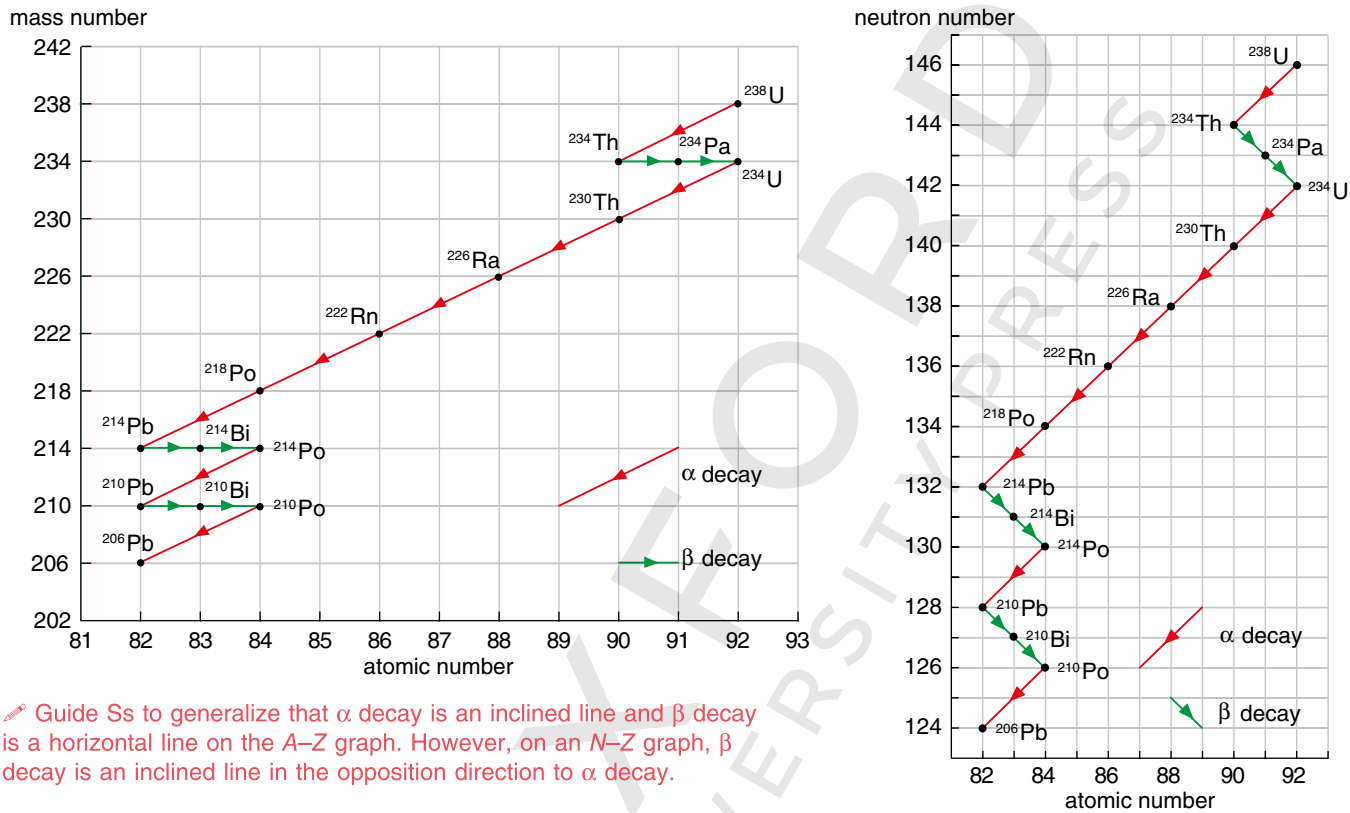
$$^{238}_{92}\text{U} \xrightarrow{\alpha} {}^{234}_{90}\text{Th} \xrightarrow{\beta} {}^{234}_{91}\text{Pa} \xrightarrow{\beta} {}^{234}_{92}\text{U} \longrightarrow \cdots \longrightarrow {}^{206}_{82}\text{Pb}$$

The series involves a number of decays and some intermediate ones are not shown in the equation.

decay series 衰變系

In Figure 2.2e, Bi-214 may also undergo an α decay to become Tl-210 and then a β decay to become Pb-210. The diagram only shows a branch of the decay series.

We may also represent the decay series graphically (Fig 2.2e). Note that we can identify the type of decay according to the orientation of the arrow on the graph. γ emission cannot be shown in these diagrams because nuclei do not change their composition in this kind of decay.



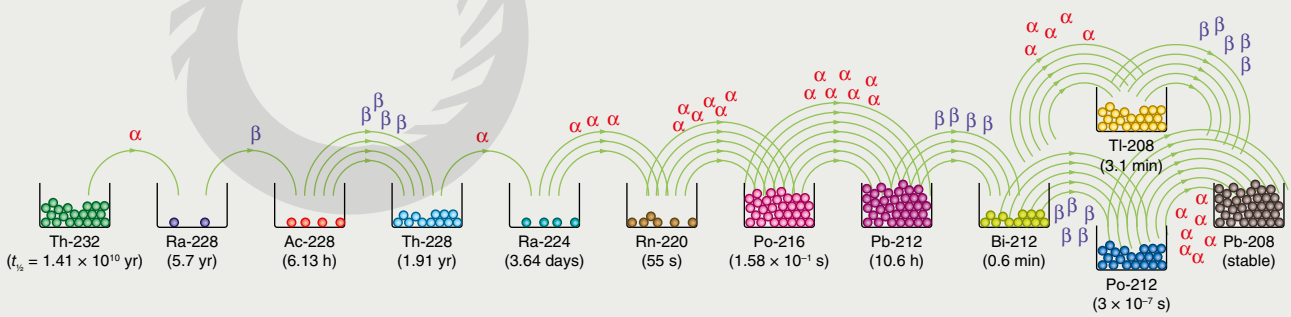
Guide Ss to generalize that α decay is an inclined line and β decay is a horizontal line on the *A*–*Z* graph. However, on an *N*–*Z* graph, β decay is an inclined line in the opposition direction to α decay.

Fig 2.2e *Decay series of uranium-238 by using A–Z and N–Z diagrams.*

Know more

Natural decay series

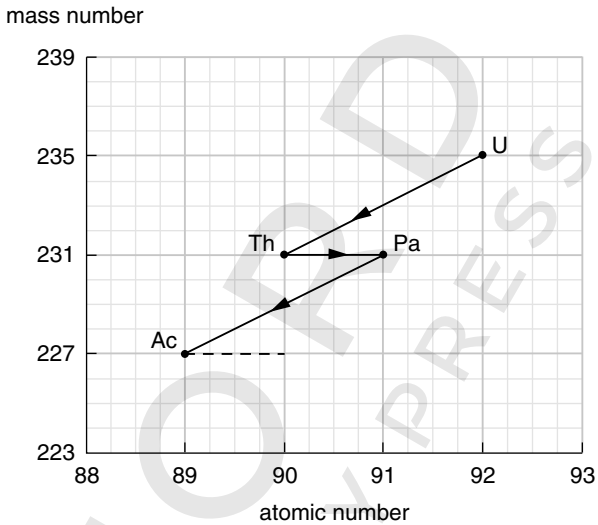
There are three natural decay series, with initial nuclides U-235, U-238 and Th-232. They end up as lead isotopes (Pb-207, Pb-206 and Pb-208 respectively) via α and β decays. The U-238 decay series is shown in Figure 2.2e above. The figure below shows an artist illustration of the Th-232 decay series.



Artist illustration of the Th-232 decay series ($t_{½}$ shown will be discussed later).

Example 3 | Decay series of uranium-235

The decay series of uranium-235 is shown below.



(a) What kind of particle is emitted in each decay shown above?

(b) The decay series ends with a stable lead-207 ($^{207}_{82}$Pb). How many α particles and β particles are emitted in the complete series?

(c) From the above figure, can you deduce whether γ radiation is emitted in the decay series? Explain your answer.

Solution

(a) $^{235}_{92}\text{U} \xrightarrow{\alpha} {}^{231}_{90}\text{Th} \xrightarrow{\beta} {}^{231}_{91}\text{Pa} \xrightarrow{\alpha} {}^{227}_{89}\text{Ac} \longrightarrow \cdots$

Tip
The mass number changes in an α decay, but not in a β decay or γ emission.

(b) Number of α particles emitted

$$= \frac{\text{change in mass number}}{\text{mass number of } \alpha \text{ particle}}$$

$$= \frac{235 - 207}{4}$$

$$= 7$$

The decay series can be written as:

$$^{235}_{92}\text{U} \longrightarrow {}^{207}_{82}\text{Pb} + 7\,{}^{4}_{2}\text{He} + y\,{}^{0}_{-1}\text{e}$$

Consider the atomic numbers.

$$92 = 82 + 7 \times 2 + y \times (-1)$$

$$y = 4$$

Four β particles are emitted in the complete series.

(c) No. γ emission does not involve any change in the atomic number and the mass number. Therefore, it cannot be shown on the graph.

Learn by practice 2

1 A polonium-216 nucleus decays into a lead-212 nucleus and decays again into a bismuth-212 nucleus.

$$^{216}_{84}\text{Po} \xrightarrow{\alpha} {}^{212}_{82}\text{Pb} \xrightarrow{\beta} {}^{212}_{83}\text{Bi}$$

Draw the graphical representations of the above decays in Figures a and b.

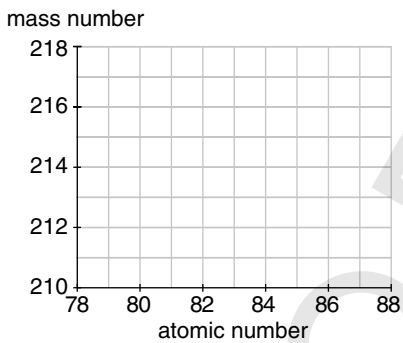


Fig a

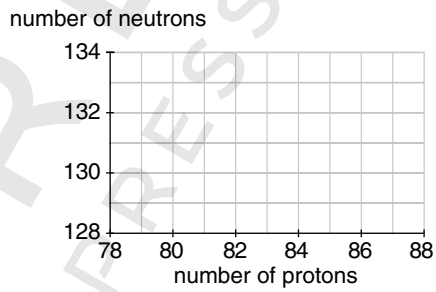


Fig b

2 Find the number of α particles and β particles emitted in each of the following decay series.

(a) From $^{238}_{92}\text{U}$ to $^{230}_{90}\text{Th}$

(b) From $^{239}_{94}\text{Pu}$ to $^{207}_{82}\text{Pb}$

(c) From $^{226}_{88}\text{Ra}$ to $^{206}_{82}\text{Pb}$

☆ HKDSE 2020 Paper 1A Q31

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Number of protons and neutrons

Nuclide P undergoes two radioactive decays to form nuclide R as shown.

$$P \xrightarrow{\alpha} Q \xrightarrow{\beta} R$$

Which of the following correctly describes the numbers of protons and neutrons in P and R?

A P has 2 more protons than R.

B P has 3 more protons than R.

C P has 3 more neutrons than R.

D P has 4 more neutrons than R.

Solution

The decay series can be rewritten as:

$$^{Y}_{X}P \xrightarrow{\alpha} {}^{Y-4}_{X-2}Q \xrightarrow{\beta} {}^{Y-4}_{X-1}R$$

Assign an atomic number and a mass number for the initial nuclide and work out those for the remaining nuclides.

The numbers of protons in P and in R are X and $X - 1$ respectively.

$\therefore$ A and B are incorrect.

Number of neutrons in $P = Y - X$

Number of neutrons = $A - Z$

Number of neutrons in $R = (Y - 4) - (X - 1) = Y - X - 3$

$\therefore$ The answer is C.

▶ Revision 2 Q14 (p.84)

DSE goal

Examine the random nature of radioactive decay.

State the proportional relationship between the activity of a sample and the number of undecayed nuclei.

DSE exam

20(1A)Q30

3 Decay curve

Dice throwing is a random process. When you throw many dice, you cannot predict which dice will show a '1' after a particular throw nor when a particular dice, e.g. the yellow one (Fig 2.2f), will show '1'. However, if you throw many dice at the same time, you can roughly estimate how many dice will land on a '1'! Radioactive decay is also a random process and can be studied using an analogy with dice.



Fig 2.2f *Dice throwing is a random process.*

→ This video demonstrates Expt 2a.

Video

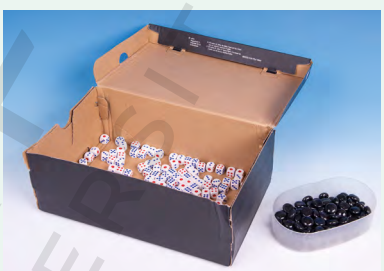
A large number of dice should be used to produce accurate statistical results. ▶

By working in groups and summing up the results, the effect of sample size could be demonstrated. A more accurate decay curve will be shown.

→ This is a Video Quiz about this experiment.

Experiment 2a Radioactive decay — a dice analogy

1 Place 100 dice into a large shoe box. Cover the box.



2 Throw the dice by shaking the box. Open the box. Count the number of dice with side '1' facing up and replace them with black buttons.

3 Repeat step 2 until only a few dice remain in the box. Plot a graph of the number of dice remaining against the number of throws.

Discussion

1 Compare your results with other groups. Are they the same?

2 From the graph, find the number of throws needed to reduce the number of dice from 100 to 50 and from 50 to 25. What do you discover?

The above experiment demonstrates a dice analogy which can simulate radioactive decay in the following way:

Dice analogy		Radioactive decay
Dice	represents	radioactive nucleus
Dice showing '1'		decaying nucleus
Black button		daughter nucleus
Turn of throwing		fixed period of time

Table 2.2c *The dice analogy and the radioactive decay.*

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We should note the following from the experimental results:

1 Different groups of students would get slightly different results because of the random nature of dice throwing.

2 In general, the number of dice showing '1' decreases with the number of dice.

3 The number of dice decreases by half in roughly a fixed number of throws. Figure 2.2g shows the sum of 10 sets of results. The number of dice becomes halved approximately every 3.8 throws.

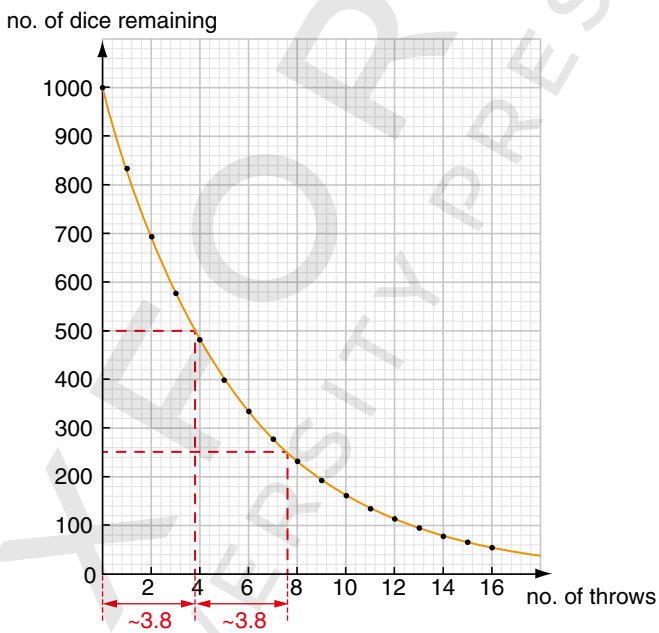


Fig 2.2g *Sample result of Experiment 2a (sum of 10 sets of results).*

Radioactive decay has similar characteristics to dice throwing. The details will be discussed in the following part.

a Random nature

Refer to point 1 above. ▶ Radioactive decay is **random in nature**. We cannot predict which nucleus will decay at a certain moment and when a particular nucleus will decay (Fig 2.2h).

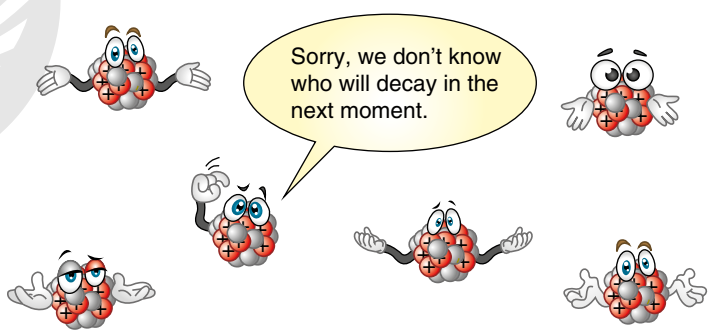


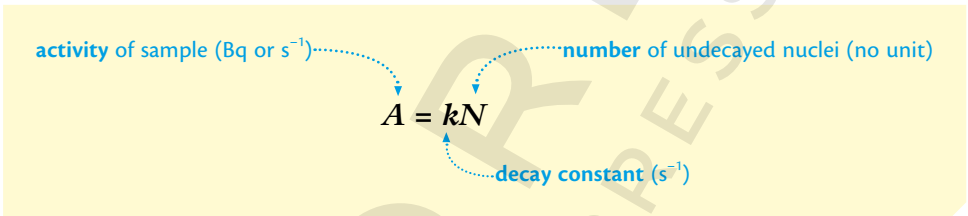
Fig 2.2h *Random nature of radioactive decay.*

b Activity

The rate of decay of a sample is called its **activity**. It is measured in **number of decays per second** (s^{-1}) or **becquerels** (**Bq**). When the activity of a radioactive sample is higher, the count rate measured by a GM counter is higher.

Refer to point 2 on p.58.

▶ The activity A of a radioactive sample is directly proportional to the number of undecayed nuclei N. This relationship can be expressed as:



✎ Ts can guide Ss that the number of dice 'decaying' each time is about $\frac{1}{6}$ of the remaining dice, i.e. the 'activity' is always directly proportional to the number of 'undecayed' dice.

We can also say that the higher the probability that a radioactive nucleus decays within a certain period of time, the larger the decay constant of the radionuclide.

▶ The **decay constant** k is usually measured in s^{-1}. Its value depends on the type of radionuclide. The larger the fraction of the radioactive nuclei decay per unit time, the larger the decay constant of the radionuclide.

c Half-life

Refer to point 3 on p.58.

DSE goal

Define half-life.

Determine the half-life of a radioisotope from its decay curve or from numerical data.

Solve problems involving radioactive decay.

DSE exam

16(1A)Q33, 18(1A)Q32, 20(1A)Q33, 21(1A)Q32, 23(1A)Q31

▶ The number of undecayed nuclei in a radioactive substance decreases by half after a fixed time interval. This time interval is called **half-life**, $t_{½}$. It is also the time taken for the activity to reduce by half.

The value of half-lives depends on the type of radionuclide. Table 2.2d gives the half-lives of some common radionuclides. Note that the half-life is a characteristic of the radionuclide. It does not vary over time and will not change even if there are changes in external conditions such as temperature and pressure.

It is useful to know the half-life of a radionuclide because

1 we can find out how long a sample will remain radioactive and hazardous;

2 we can use it to identify unknown radionuclides since each has a unique half-life.

Radionuclide	Half-life	Radionuclide	Half-life
Uranium-235	7×10^8 years	Iodine-131	8 days
Carbon-14	5730 years	Sodium-24	15 hours
Radium-226	1620 years	Radon-220	56 s
Caesium-137	30 years	Helium-5	7.6×10^{-22} s

Table 2.2d *Half-lives of several radionuclides.*

activity 放射強度　becquerel (Bq) 貝克　decay constant 衰變常數　half-life 半衰期

The half-life of a radioactive substance is the time over which the number of radioactive nuclei decreases by half. It is also equal to the time taken for the activity of a radioactive sample to fall by half.

Let us take iodine-131 (I-131) as an example. We can find the value of its half-life from its **decay curve** (Fig 2.2i).

Simulation

→ This simulation shows the decay process of radioactive nuclei. Once the simulation starts, the numbers of decayed and undecayed nuclei present at any time are displayed. Finally, a decay curve is plotted and the half-life is marked on the curve.

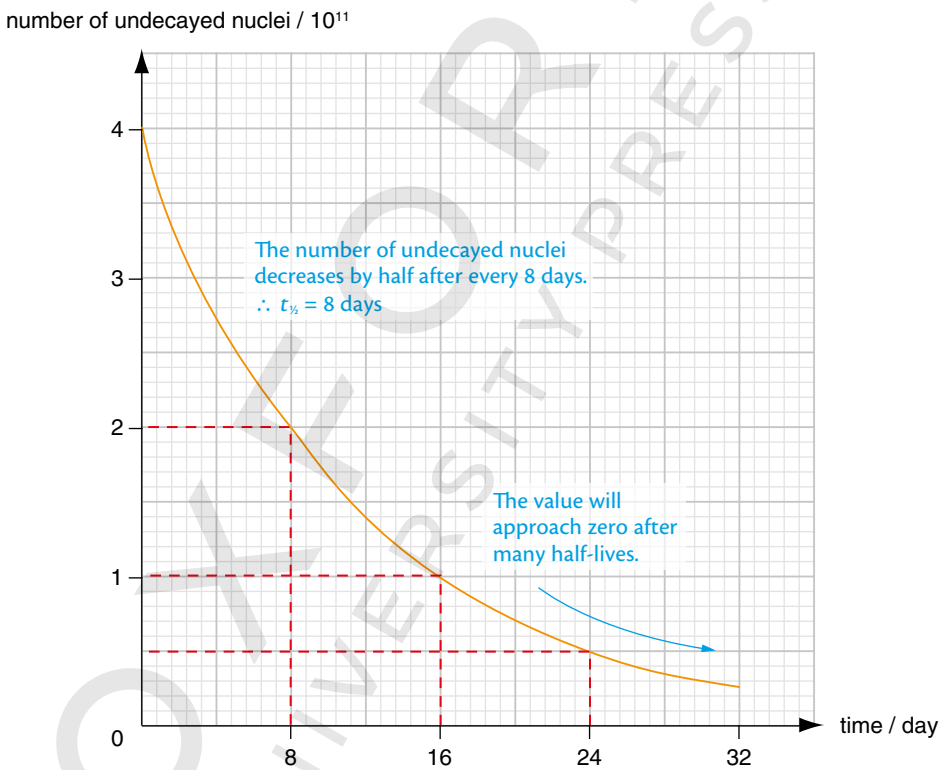


Fig 2.2i *Finding half-life from the decay curve of an iodine-131 sample (by drawing red dotted lines).*

The number of undecayed nuclei decreases from 4×10^{11} to 2×10^{11} after 8 days, it decreases from 2×10^{11} to 10^{11} after another 8 days, and so on. Therefore, we can see that the half-life of I-131 is 8 days. The same result can be obtained if the y-axis represents the activity of the I-131 sample.

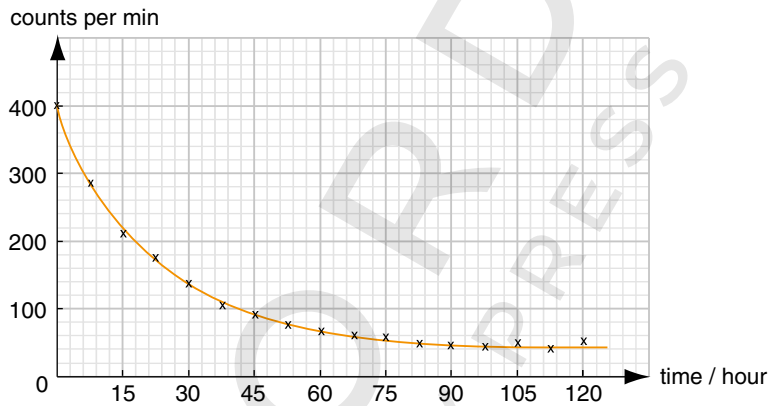
It is worth noting that:

- The total number of nuclei in the sample may or may not change over time. The decayed nuclei may still remain in the sample.
- We have no idea when a particular radioactive nucleus will decay since radioactive decay is a random process. However, if there are a huge number of radioactive nuclei, we can predict how many nuclei will decay within a certain period of time according to the decay constant and the half-life.

decay curve 衰變曲線

Example 4 | Finding half-life from count rate

A GM counter is used to measure the count rate of a sodium-24 sample over a period of time. The graph below shows how the count rate varies with time. Sodium-24 will decay to form a stable product.



(a) Why does the graph flatten out rather than fall to zero after some time?

(b) Estimate the half-life of sodium-24 from the graph.

Solution

(a) In that period, the activity of the sample falls to almost zero. The count rate measured by the GM counter is due to background radiation.

(b) Count rate due to background radiation = 40 counts per min

Initial count rate due to sample = 400 – 40 = 360 counts per min

When $t = t_{½}$,

measured count rate = $\frac{360}{2}$ + 40 = 220 counts per min

From the graph,

$t_{½}$ = 15 h

→ Teaching notes and exercises are available on OUP web.

Skill
Problem-solving

Finding half-life from count rate

① Find the count rate due to background radiation.

② Calculate the initial count rate due to the sample.

③ Find the count rate due to the sample after one $t_{½}$, i.e. the count rate is reduced by half.

④ Add the values of ① and ③. This gives the measured count rate after one $t_{½}$.

⑤ Read from the graph to see how long the measured count rate takes to change from the initial value to the value in ④. This equals $t_{½}$.

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Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* It is impossible to predict when a particular nucleus will decay. (T/F)

2 *True or false:* It is impossible to predict which nuclei will decay at a particular moment. (T/F)

3 A radioactive sample has 10^{10} undecayed nuclei and a decay constant of 0.01 s^{-1}. Find the activity of the sample.

4 The activity of a pure β source is 500 Bq and it has a half-life of several days. Are the following statements correct? Put a tick in the corresponding box(es).

☐ About 500 nuclei will decay in the next second.

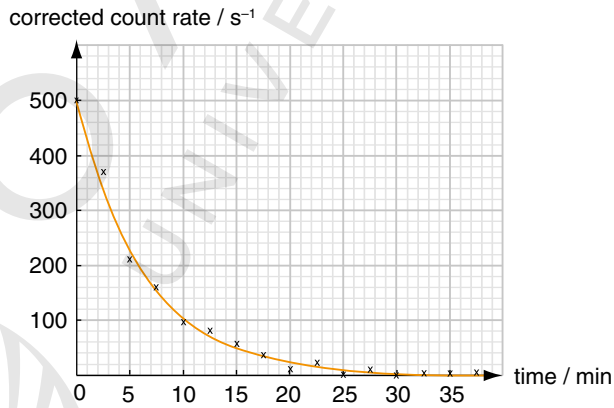
☐ The source will emit about 500 β particles in the next second.

5 The activity of a radioactive sample is 0.02 Bq. If the decay constant of the radioactive nuclei in the sample is 4.8×10^{-7} s^{-1}, find the number of undecayed nuclei in the sample.

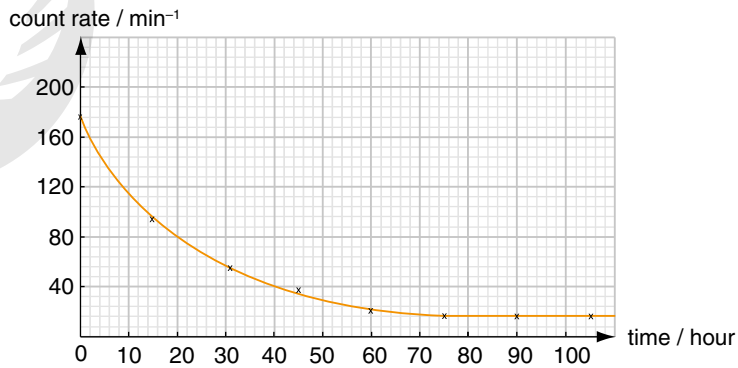
6 *True or false:* Each radionuclide has a unique half-life. (T/F)

7 In each of the following cases, a radioactive source is placed in front of a GM counter and a graph is plotted. Estimate the half-life of the radioactive source in each case.

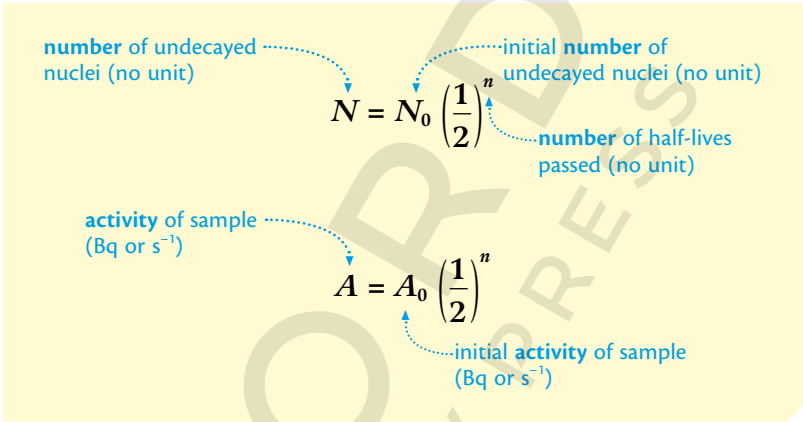
(a)



(b)



When n half-lives pass, the number of undecayed nuclei and the activity will both be halved by n times and decrease to $\left(\frac{1}{2}\right)^n$ of their initial values. Suppose that N_0 is the initial number of undecayed nuclei and A_0 is the initial activity of the sample, then



Example 5 Activity and half-life

Sodium-24 will decay to form a stable product. It has a decay constant of 1.28×10^{-5} s^{-1} and a half-life of 15 hours. A sample of sodium-24 has an activity of 8×10^5 Bq.

(a) Find the number of undecayed nuclei in the sample.

(b) What will be the activity of the sample after 45 hours?

(c) Find the time needed for the activity to decrease to 2.5×10^4 Bq.

Solution

(a) By $A = kN$,

$$N = \frac{A}{k} = \frac{8 \times 10^5}{1.28 \times 10^{-5}} = 6.25 \times 10^{10}$$

(b) Number of half-lives passed $= \frac{45}{15} = 3$

$$A = A_0\left(\frac{1}{2}\right)^n = (8 \times 10^5) \times \left(\frac{1}{2}\right)^3 = 1 \times 10^5 \text{ Bq}$$

(c) 8×10^5 Bq $\xrightarrow{\times\frac{1}{2}}$ 4×10^5 Bq $\xrightarrow{\times\frac{1}{2}}$ 2×10^5 Bq $\xrightarrow{\times\frac{1}{2}}$ 1×10^5 Bq $\xrightarrow{\times\frac{1}{2}}$ 5×10^4 Bq $\xrightarrow{\times\frac{1}{2}}$ 2.5×10^4 Bq

The activity is halved by 5 times and therefore 5 half-lives are needed.

$\therefore$ Time needed $= 15 \times 5 = 75$ h

Alternative method:

By $A = A_0\left(\frac{1}{2}\right)^n$,

$$\left(\frac{1}{2}\right)^n = \frac{A}{A_0}$$

$$= \frac{2.5 \times 10^4}{8 \times 10^5}$$

$$= \frac{1}{32}$$

$$= \left(\frac{1}{2}\right)^5$$

$\therefore$ 5 half-lives are needed.

Time needed $= 15 \times 5 = 75$ h

The change of N and A over time is concluded in Figure 2.2j.

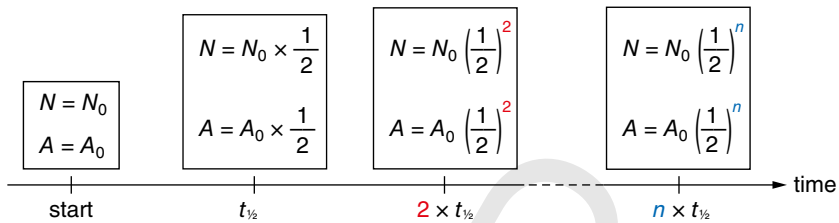


Fig 2.2j *Change of N and A over time.*

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Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 Complete the following table.

	A_0 / Bq	Time passed	Half-life	Number of half-lives passed	A / Bq
(a)	50	32 h	8 h		
(b)	2		37 s		0.25
(c)		2 min	20 s		0.1

Learn by practice 3

1 Iodine-131 has a half-life of 8 days. It is known that a radioactive sample contains 16 mg of iodine-131.

(a) How many half-lives have passed in 24 days?

(b) What is the mass of iodine-131 in the sample after 24 days?

2 The initial activity of a radioactive sample is 500 kBq and its half-life is 5 hours. What is its activity after 30 hours?

3 The activity of a radioactive source drops from 600 Bq to 150 Bq in 40 hours. Find its half-life.

4 A radioactive sample contains 6×10^{12} undecayed nuclei. The half-life of the sample is 5 days. Find the number of undecayed nuclei in the sample 10 days ago.

Ext

DSE goal

Represent the number of undecayed nuclei by $N = N_0e^{-kt}$.

Apply $N = N_0e^{-kt}$ to solve problems.

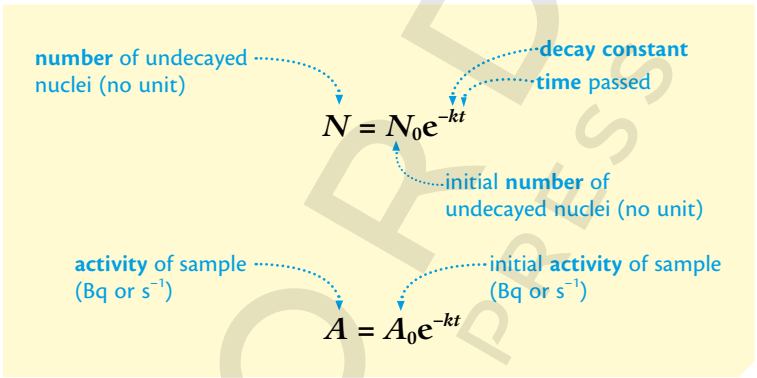
Relate the decay constant and the half-life.

DSE exam

13(1B)Q9(a), 18(1B)Q10, 19(1B)Q10(b), 20(1A)Q32, 21(1B)Q9, 22(1A)Q32, 23(1B)Q9(b), (c)(ii), etc.

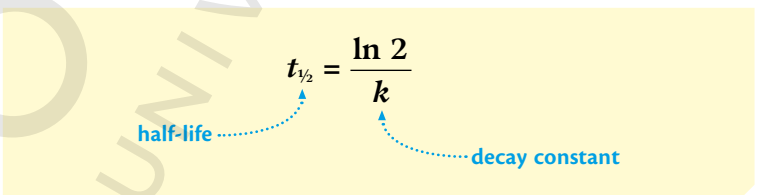
d Exponential decay

Careful analysis of radioactive decay shows that the decay curve can be described by an *exponential decay* function. The number of undecayed nuclei N and the activity A due to the nuclei vary with time t according to the following equations:



e is a constant. You can find this value on your calculator to make the calculation. In the above equations, t can be measured in any unit of time, but the units for t and k should match each other (e.g. t in min and k in min^{-1}).

From these equations, we can also find the relationship between the decay constant k and the half-life $t_{½}$. Again the units for $t_{½}$ and k should match each other in the equation.



✎ We can derive the exponential decay law as follows:

The rate of decay $\left[\frac{dN}{dt}\right]$ is proportional to the number of nuclei remaining (N). Thus we have:

$$\frac{dN}{dt} = -kN$$

$$\frac{dN}{N} = -kdt$$

Integrate both sides and substitute the initial condition:

$$\ln\left[\frac{N}{N_0}\right] = -kt$$

$$N = N_0e^{-kt}$$

When one half-life passes, the number of undecayed nuclei is reduced by half.

$$\frac{N_0}{2} = N_0\,e^{-kt_{½}}$$

$$\frac{1}{2} = e^{-kt_{½}}$$

$$\ln(2^{-1}) = -kt_{½}$$

$$t_{½} = \frac{-\ln 2}{-k} = \frac{\ln 2}{k}$$

This shows that the decay constant is also a characteristic of radionuclide. Each type of radionuclide has a unique decay constant.

✎ e is sometimes called Euler's number.

→ Teaching notes and exercises are available on OUP web.

Skill Mathematics

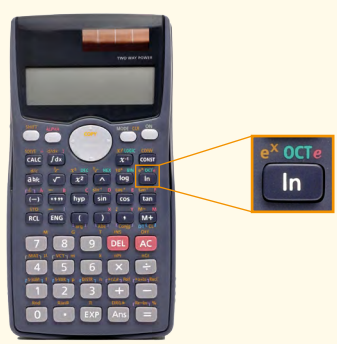
The number e and natural logarithm

The number e is a mathematical constant with a value of approximately 2.72. Like π, it is one of the most important constants in mathematics and science.

The natural logarithm (ln) is the logarithm to the base e. According to the laws of logarithms,

$\ln e = 1$ and $\ln e^x = x$

You can find the value of e (press ALPHA then e^x OCTe ln) and the natural logarithm function on your calculator to make the calculation.



exponential decay 指數式衰變

Ext

Skill *Mathematics*

Solving exponential equations with an unknown index

▸ p.121

Example 6 | Exponential decay

A sample of a radionuclide has an activity of 400 kBq. The decay constant of the radionuclide is 1.00×10^{-6} s^{-1}. The radionuclide will decay to form a stable product.

(a) Find the half-life of the sample.

(b) Find the time needed for its activity to decrease to 1 kBq.

Solution

(a) $t_{½} = \dfrac{\ln 2}{k} = \dfrac{\ln 2}{1.00 \times 10^{-6}} = 6.93 \times 10^{5}$ s

(b) By $A = A_0 e^{-kt}$,

$$\frac{A}{A_0} = e^{-kt}$$

$$\ln\left(\frac{A}{A_0}\right) = \ln e^{-kt}$$

$$\ln\left(\frac{A}{A_0}\right) = -kt$$

$$t = -\frac{1}{k}\ln\left(\frac{A}{A_0}\right) = -\frac{1}{1.00 \times 10^{-6}}\ln\left(\frac{1 \times 10^{3}}{400 \times 10^{3}}\right) = 5.99 \times 10^{6} \text{ s}$$

∴ It takes 5.99×10^{6} s for the activity to decrease to 1 kBq.

Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 Complete the following table by applying $A = A_0 e^{-kt}$.

	A_0	k	t	A
(a)		0.006 s^{-1}	20 s	0.1 Bq
(b)	2 Bq		37 min	0.25 Bq
(c)	50 Bq	0.02 h^{-1}	8 h	

2 Complete the following table by applying $t_{½} = \dfrac{\ln 2}{k}$.

	$t_{½}$	k
(a)		0.5 s^{-1}
(b)	25 s	
(c)		0.004 min^{-1}
(d)	5730 yr	

Ext

Example 7 | Decay of radium-226

Radium-226 (Ra-226) decays to radon (Rn) by emitting an α particle and has a half-life of 1600 years. The radon is also radioactive and in turn decays into other radionuclides. The process carries on until a stable lead (Pb) nuclide is formed.

(a) $^{206}_{82}$Pb, $^{207}_{82}$Pb and $^{208}_{82}$Pb are three stable isotopes of lead. Determine which isotope(s) could be the nuclide formed in the above decay series.

(b) A sample contains 1 mg of Ra-226.

(i) Show that the mass of undecayed Ra-226 will be about 0.98 mg after 50 years.

(ii) Simon says, 'The mass of the sample will be reduced to 0.98 mg after 50 years.' Is he correct? Explain.

(iii) The corrected count rate of the sample will not be 98% of the original value after 50 years. Explain the reason.

Solution

Tip
The mass number is reduced by 4 during an α decay and remains unchanged during a β decay.

(a) The change in mass number in a decay series must be a multiple of 4.

∴ Only $^{206}_{82}$Pb can be the nuclide formed in the decay series.

(b) (i) $k = \frac{\ln 2}{t_{½}} = \frac{\ln 2}{1600} = 4.33 \times 10^{-4}\ \text{yr}^{-1}$

Tip
Year and year^{-1} (yr^{-1}) are used as the units of t and k respectively.

Since mass of undecayed nuclei $m \propto N$ and $N = N_0 e^{-kt}$,

$m = m_0 e^{-kt} = 1 \times e^{-(4.33 \times 10^{-4}) \times 50} = 0.979\ \text{mg} \approx 0.98\ \text{mg}$

Alternative method:

Since mass of undecayed nuclei $m \propto N$ and $N = N_0\left(\frac{1}{2}\right)^n$,

Tip
In applying this equation, the number of half-lives n needs not be an integer and can be expressed as $\frac{t}{t_{½}}$.

$m = m_0\left(\frac{1}{2}\right)^n$

$= 1 \times \left(\frac{1}{2}\right)^{\frac{50}{1600}}$

$= 0.979\ \text{mg} \approx 0.98\ \text{mg}$

(ii) He is wrong.

Although 0.02 mg of Ra-226 decay in 50 years, the decayed nuclei do not disappear but instead change into other nuclei which also make up the mass of the sample.

(iii) Most products in the decay series, including radon, are radioactive and contribute to the measured count rate.

Radon is a gas which may dissipate in air. However, it has a short half-life (~3.8 days) and its subsequent decay products may deposit back to the sample.

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Ext

Learn by practice 4

Assume all decay products are stable in the following questions.

1 The decay constant of radionuclide X is 1.28×10^{-5} s^{-1}.

(a) Find the half-life of X.

(b) The number of undecayed nuclei in a sample of X is 3×10^{10}. What is the number of undecayed nuclei after 1 day?

2 A sample of a radionuclide has an activity of 3.2×10^{4} Bq. The decay constant of the radionuclide is 0.578 min^{-1}. What is the activity of the sample 3 min ago?

3 The activity of a sample of radionuclide drops from 3.2×10^{4} Bq to 3.8×10^{3} Bq in 1.5 h. Find the decay constant of the radionuclide.

4 The decay constant of a radionuclide is 2.41×10^{-5} s^{-1}. How long will it take for the activity of a sample of this radionuclide to drop from 7.4×10^{3} Bq to 5.2×10^{2} Bq?

5 Cobalt-60 has a half-life of 5.27 years. The initial activity of a cobalt-60 sample is 7.4×10^{4} Bq. Find the activity of the sample after 2 years.

6 Sodium-24 has a half-life of 15 hours. How long will it take for the activity of a sodium-24 sample to drop from 100 000 Bq to 10 000 Bq?

7 A sample of radium-224 has an initial activity of 3.00×10^{5} Bq. After 10 days, its activity becomes 4.45×10^{4} Bq. Find the half-life of radium-224.

Know more

Curie

Curie (Ci) is another unit of activity. It is defined as the activity of one gram of radium and is equivalent to about 3.7×10^{10} Bq. It had been used for many years as the standard unit for measuring activity. It was named in honour of Pierre Curie and his wife Marie Curie.



Marie Curie (left) and Pierre Curie (right).

OUP eResources

Exercise 2.2

1 Nucleus X is formed when $^{241}_{95}$Am undergoes an α decay, and $^{238}_{94}$Pu is formed when nucleus Y undergoes a β decay. Which of the following representations for X and Y is correct?

	X	Y
A	$^{237}_{93}X$	$^{238}_{93}Y$
B	$^{237}_{93}X$	$^{238}_{95}Y$
C	$^{237}_{93}X$	$^{237}_{94}Y$
D	$^{245}_{97}X$	$^{238}_{95}Y$

★**2** The activity of a β source is 500 Bq and it has a decay constant of 10^{-6} s^{-1}. Which of the following statements is **incorrect**?

A About 500 nuclei will decay in the next second.

B The source will emit about 500 β particles in the next second.

C About 10^{-6} of the nuclei in the source are unstable.

D All of the above.

★**3** In an experiment, a radioactive substance is placed in front of a GM counter and the change in count rate over time is plotted (Fig a).

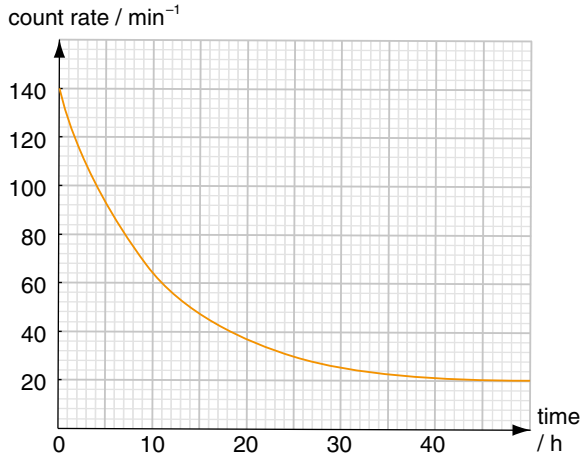


Fig a

Which of the following statements are correct?

(1) The background count rate is 20 min^{-1}.

(2) The half-life of the radioactive substance is 7 hours.

(3) If the experiment started one half-life earlier, the initial count rate would be 260 min^{-1}.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★**4** The decay series of plutonium-239 is shown by the following equation.

$$^{239}_{94}\text{Pu} \xrightarrow{\text{(i)}} {}^{235}_{92}\text{U} \xrightarrow{\text{(ii)}} {}^{231}_{90}\text{Th} \xrightarrow{\text{(iii)}} {}^{231}_{91}\text{Pa} \longrightarrow \cdots \longrightarrow {}^{207}_{82}\text{Pb}$$

(a) What kind of particle is emitted in decay processes (i), (ii) and (iii) respectively?

(b) Find the number of α particles and the number of β particles emitted in the entire series.

★**5** Radionuclide *X* undergoes a β decay to form nuclide *Y*-137 (atomic number 56).

(a) Write an equation to represent the above decay process.

(b) γ rays are emitted after the above decay process.

(i) Why are gamma rays emitted?

(ii) A decay series can be represented in a graph of mass number against atomic number. Can the γ emission be shown on the graph? Why?

★**6** A GM counter is set up and records a background count rate of 70 min^{-1}. When a radioactive source is placed in front of the GM tube, the count rate increases to 310 min^{-1}. Predict the count rate recorded by the GM counter after two half-lives of the source.

★**7** A nuclide *A* undergoes a series of decays to form nuclide *F* (Fig b).

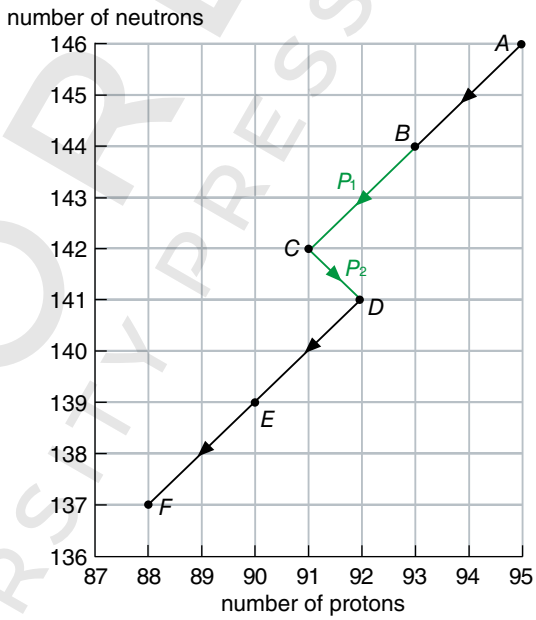


Fig b

(a) What kinds of particles are emitted during the radioactive decays P_1 and P_2?

(b) Write an equation for the decay from *E* to *F*.

(c) Are there isotopes of the same element in the decay series? If so, name the isotopes.

★**8** Ext A radioactive sample has an activity of 4×10^3 Bq. Its half-life is 4 hours.

(a) Find the number of undecayed nuclei in the sample.

(b) Find the number of undecayed nuclei in the sample after 2 hours.

★**9** Ext The number of undecayed nuclei in a radioactive sample drops by $\frac{1}{3}$ in 1 hour.

(a) Find the decay constant of the sample.

(b) If the initial activity of the sample is A_0, what is the activity of the sample

(i) after 1 hour,

(ii) after 2 hours?

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2.3 Uses of radioactive isotopes

Nowadays, radionuclides are used in many products and technologies. The choice of radionuclides depends on their characteristics and the type of radiation they emit. In this part, we will learn how the ionizing power, penetrating power and half-lives of radionuclides are utilized in different applications.

DSE goal

Discuss uses of radioactive isotopes.

DSE exam

13(1B)Q9, 15(1A)Q33, 16(1B)Q9, 17(1B)Q10

1 Ionizing power

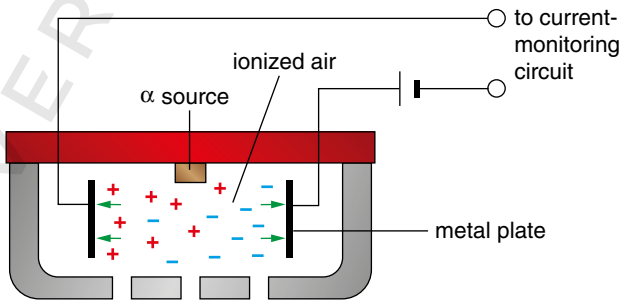
Some appliances work by relying on the production of ions. Since **α radiation has the strongest ionizing power**, α sources are suitable for the purposes of these appliances (Fig 2.3a).

Ionization smoke detector

The α source used usually has a long half-life so that the level of ionization can remain stable over the lifetime of the smoke detector.

α particles emitted from the source ionize air molecules and create a flow of ions (charged particles) between two charged metal plates. As a result, a constant current flows between the two plates under normal conditions.
When smoke enters the detector, the path of some ions are blocked. As a result, the current drops and the ring alarm is triggered.

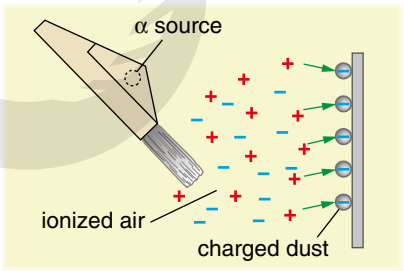




Anti-static brush

Since α radiation has a range of only a few centimetres in air, the α source is fixed near the brush hair.

α particles emitted from the source in an anti-static brush ionize air molecules near its surface. The ions produced then neutralize the charges on the dust, making it easier to be removed.



Dust may easily adhere to the surface of photos or film due to electrostatic attraction.



Fig 2.3a *Ionization smoke detector and anti-static brush.*

2 Penetrating power

γ and β radiations have stronger penetrating power than α radiation. γ and β sources are therefore used in appliances that need to penetrate objects.

a Radioactive tracers

Radioactive tracers made of a γ source can be detected outside a structure. The distribution of the tracer or the path of its flow can therefore be monitored from outside. Radioactive tracers can be used in medicine (Fig 2.3b) and in environmental protection (Fig 2.3c).

Radionuclide imaging
A weak γ source with a short half-life of several hours (e.g. technetium-99m or iodine-123) is injected into the body of a patient. After some time, a gamma camera is used to view the distribution of the radionuclides inside the body. This technique can reveal abnormalities in organs.

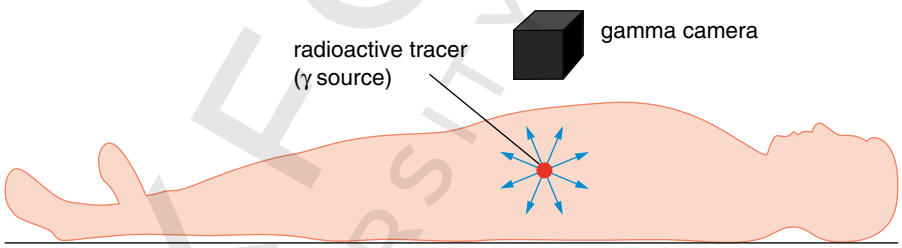


Fig 2.3b Medical imaging using a radioactive tracer.

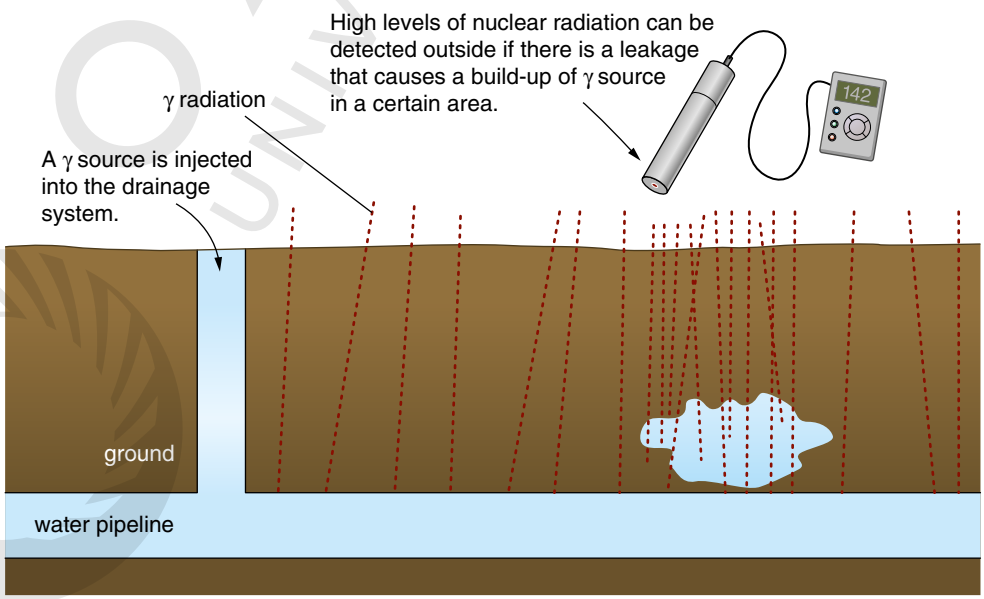


Fig 2.3c Using a radioactive tracer to detect the leak in a underground drainage system.

The source must have a half-life long enough for engineers to detect leakage but short enough to reduce the long-term effect on the environment.

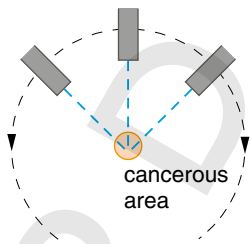
The effect of the radiation is not dangerous at these very low doses. In addition, these radioisotopes have a short half-life (from a few mintues to a few days) so they quickly vanish from the bodies or the environment.

b Radiotherapy

Many diseases such as cancer can be cured by *radiotherapy*. Figure 2.3d shows how it works.

✎ α source coupled to tumour selective carrier molecules can also be used to kill cancer cells. As the range of α radiation is short, it causes less damage to nearby healthy tissue. For detail, refer to the following website.





A γ source like cabalt-60 is made to rotate around the cancerous area. A highly concentrated beam is emitted from the source to destroy the cancer cells. The surrounding healthy cells receive much less radiation and less damage is caused.

Fig 2.3d *Radiotherapy.*

c Sterilization of food and medical instruments

Food can be sterilized when exposed to γ radiation. This process is called food irradiation. The radiation can kill bacteria, mould and insects in food (Fig 2.3e and 2.3f). This can prolong the shelf life of the food. Note that the irradiated food is exposed to ionizing radiation but does not become radioactive itself. Medical instruments can also be sterilized in a similar way.

To learn more about the safety of irradiated food, read the report issued by the Centre for Food Safety.

Useful link



Fig 2.3e *Labelling is required for irradiated food in Hong Kong.*



Fig 2.3f *International food irradiation symbol.*

d Thickness gauge

A thickness gauge is commonly used for paper milling and the production of aluminium foil in industry. Figure 2.3g shows how it works.

A thin aluminium foil is dragged between a β source and a detector. If the foil is too thick, the amount of radiation received by the detector will be less than usual and a signal will be sent to make the foil thinner.

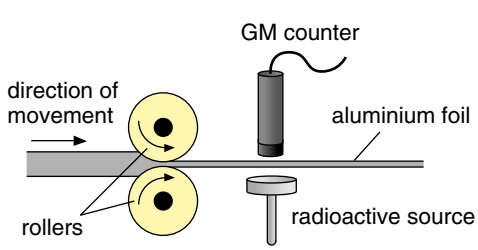
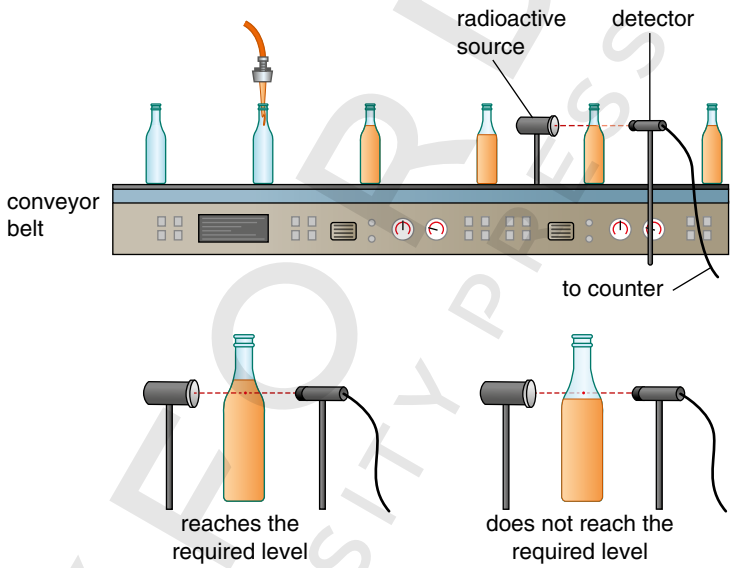


Fig 2.3g *Simplified diagram of a thickness gauge.*

radiotherapy 放射療法

Example 8 Filling bottles of juice

Fruit juices contained in plastic bottles are made in a factory. A radioactive source is placed on one side of a conveyor belt and a detector is placed at the same level on the other side. They are used to monitor whether the juice reaches the required level in a bottle.



(a) Explain how this system can detect the bottles not filled up to the required level.

(b) Which type of radioactive source (α, β or γ) should be used? Explain why the other two are not suitable.

(c) Two sources emitting the type of radiation in (b) are available. Their half-lives are 20 minutes and 4 years. Which source should be used? Explain why the other one is not suitable.

Solution

(a) Some radiation will be absorbed by the juice if the juice reaches the required level in a bottle. Therefore, if a bottle is not filled up to the required level, more radiation will pass through the bottle and the counter will record a higher count rate.

(b) A β source should be used.

An α source is not suitable because α radiation cannot pass through the bottle.

A γ source is not suitable because the effect of the juice on the penetration of γ radiation is negligible.

(c) The source with a half-life of 4 years should be used.

The source with a half-life of 20 minutes cannot maintain a stable activity for a reasonable period of time (e.g. several months).

3 Other applications

Archaeologists have benefited greatly from the use of radiometric-dating techniques. They are based on the principle that a particular radioisotope decays with a constant half-life. Using such techniques, investigators are able to determine the ages of various remains and rocks.

a Carbon-14 dating

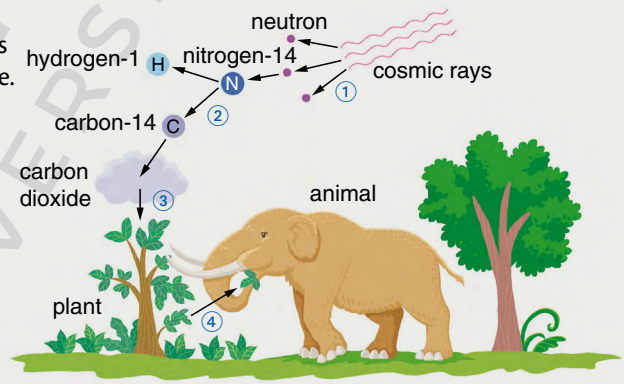
Carbon-14 dating helps determine the age of ancient remains of an organism that was once alive. This is achieved by measuring the activity of carbon atoms in the remains.

Carbon-14 (C-14) is radioactive. It emits β radiation and has a half-life of 5730 years. The concentration of C-14 is almost constant in the atmosphere. As living organisms keep absorbing and releasing carbon, the concentrations of C-14 in them are the same as that in the atmosphere.

Know more

How living organisms obtain C-14

① Cosmic rays produce neutrons upon entering the atmosphere.
② The neutrons bombard with nitrogen-14 nuclei in the atmosphere, producing C-14 nuclei ($^{14}_{7}N + {}^{1}_{0}n \rightarrow {}^{14}_{6}C + {}^{1}_{1}H$).
③ Plants take in C-14 from the atmosphere.
④ Animals take in C-14 from plants.



In a living organism, the amount of C-14 in each gram of carbon is 1.6×10^{-12} g. Due to this small ratio, in our calculations we can assume that the total amount of carbon does not change over time (see Example 9 on p.75).

When an organism dies, no extra carbon is taken in while the concentration of C-14 in the body continues to decrease due to radioactive decay (Fig 2.3h). By measuring the activity of C-14 in ancient remains, their age can be estimated.

Carbon-14 dating can be used to estimate the age of ancient remains from several hundred years to about 60 000 years ago.

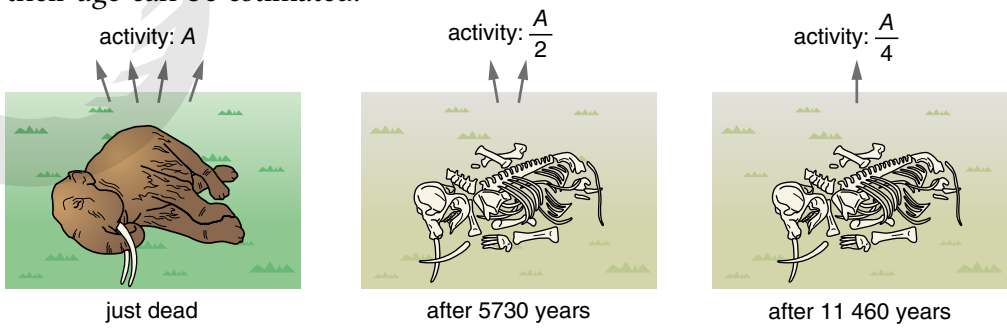


Fig 2.3h *The concentration of C-14, and hence the activity, decreases over time after the organism dies.*

carbon-14 dating 碳-14年代测定法

→ Teaching notes and exercises are available on OUP web.

Skill
Mathematics

Mass, molar mass, number of moles and number of particles

Recall that the molar mass of a kind of particle is the mass of 1 mole of that kind of particle. In addition, the Avogadro's number N_A is the number of particles in 1 mole. The following equations can therefore be used to solve problems.

- number of moles = $\frac{\text{total mass}}{\text{molar mass}}$
- number of particles = number of moles × N_A

Ext

Example 9 | Carbon-14 dating

In an archaeological site, a sample of charcoal containing 0.5 kg of carbon is found and it came from a plant which died a long time ago. The ratio of mass of carbon-14 to all carbon nuclei in a living plant is 1.6×10^{-12}. The half-life of C-14 is 5730 years. The molar mass of C-14 is 14 g.
Given: 1 year = 3.16×10^7 s, $N_A = 6.02 \times 10^{23}$ mol^{-1}

(a) A sample from a living plant contains 0.5 kg of carbon. Estimate the mass of C-14 in the sample.

(b) Estimate the activity of the sample in (a).

(c) The activity of the sample of charcoal is 11.8 Bq. Estimate the age of this sample in years.

Solution

(a) Mass of C-14 in living sample = $0.5 \times (1.6 \times 10^{-12}) = 8 \times 10^{-13}$ kg

(b) $k = \frac{\ln 2}{t_{\frac{1}{2}}} = \frac{\ln 2}{5730 \times (3.16 \times 10^7)} = 3.828 \times 10^{-12}$ s^{-1}

Number of mole $n = \frac{\text{total mass}}{\text{molar mass}} = \frac{8 \times 10^{-13}}{14 \times 10^{-3}} = 5.71 \times 10^{-11}$ mol

$N = nN_A = (5.71 \times 10^{-11}) \times (6.02 \times 10^{23}) = 3.44 \times 10^{13}$

$A = kN = (3.828 \times 10^{-12}) \times (3.44 \times 10^{13}) = 131.7$ Bq ≈ 132 Bq

(c) $k = \frac{\ln 2}{t_{\frac{1}{2}}} = \frac{\ln 2}{5730} = 1.21 \times 10^{-4}$ yr^{-1}

By $A = A_0 e^{-kt}$,

$t = -\frac{1}{k}\ln\left(\frac{A}{A_0}\right) = -\frac{1}{1.21 \times 10^{-4}}\ln\left(\frac{11.8}{131.7}\right) = 19\ 900$ yr

∴ The age of this sample is 19 900 yr.

Tip
See the steps in Example 6 on p.66. ▶

Tip
When k is in yr^{-1}, t is in yr. ▶

Think about...

Why can't carbon-14 dating be used to determine the age of remains which are over 1 million years old?

b Geological dating

Have you ever been to the Hong Kong UNESCO Global Geopark? You may see a lot of fascinating rocks there (Fig 2.3i). Geologists believe that most of these rocks were formed from volcanic activity 140 million years ago.

How do geologists estimate the age of rocks? The following DSE practice demonstrates one of the methods.



Fig 2.3i *The rocks in Sai Kung East are estimated to be formed about 140 million years ago.*

Ext

☆ HKDSE 2016 Paper 1B Q9

→ This video demonstrates how this question is solved.

DSE practice 2 | Uranium-lead dating

Video

Uranium-238 (U-238) undergoes a series of radioactive decays to form stable lead-206 (Pb-206). The decay process has an overall decay constant of $4.92 \times 10^{-18}\ s^{-1}$.

In a rock sample, the ratio of the number of U-238 nuclei to the number of Pb-206 nuclei is 4 : 1. Assume that the rock contained no Pb-206 nuclei when it was just formed, and that the amount of intermediate decay products in the sample is negligible. Estimate the age of the rock.

Take 1 year = 3.16×10^7 s.

A 1.44×10^9 yr B 1.86×10^9 yr

C 8.92×10^9 yr D 1.04×10^{10} yr

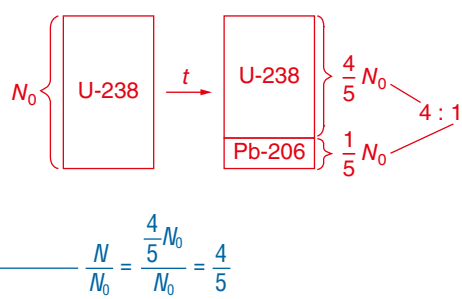
Solution

Let N_0 be the number of U-238 nuclei when the rock was just formed.

From that time to now, the total number of U-238 and Pb-206 nuclei does not change. $\frac{1}{5}$ of the U-238 nuclei undergo the decay series to form Pb-206 nuclei and $\frac{4}{5}$ of the U-238 nuclei remain.

Common mistake ⚠
Students may wrongly think that the number of U-238 nuclei falls to $\frac{1}{4}$ of its initial value.

We can draw a diagram to help us understand the relationship between different quantities:



By $N = N_0 e^{-kt}$,

$$t = -\frac{1}{k}\ln\left(\frac{N}{N_0}\right) = -\frac{1}{4.92 \times 10^{-18}}\ln\left(\frac{4}{5}\right)$$
$$= 4.54 \times 10^{16}\ \text{s}$$
$$= \frac{4.54 \times 10^{16}}{3.16 \times 10^7}$$
$$= 1.44 \times 10^9\ \text{yr}$$

$\frac{N}{N_0} = \frac{\frac{4}{5}N_0}{N_0} = \frac{4}{5}$

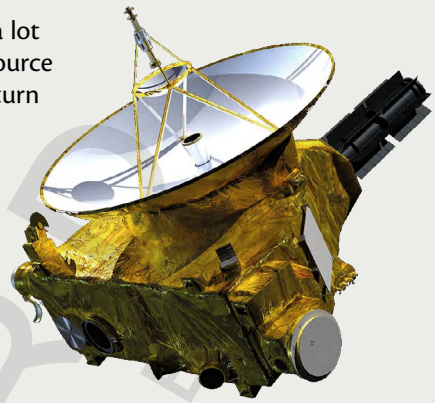
∴ The answer is A.

▶ Revision 2 Q30 (p.88)

Know more

Use of α sources to power spacecraft

An object will become hot when it absorbs a lot of nuclear radiation. In some devices, an α source is used to heat a material and the heat is in turn used to generate electricity. Unlike chemical batteries, this kind of power source can operate for decades. This makes them suitable for powering space probes and facilities in remote areas.



The New Horizons spacecraft is powered by an α source.

Google Forms Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 In ionization smoke detectors, radioactive source of (α / β / γ) radiation is used because this type of radiation has the greatest (ionizing power / penetrating power / range / half-life).

2 Some apples are exposed to gamma rays for some time before they are exported. What is the purpose of doing this?

A To kill the germs in the apples

B To neutralize the toxic substances in the apples

C To check if the apples are in good condition

3 In a thickness gauge, if the metal sheet is thicker than normal, the recorded count rate will be (higher / lower) than normal.

4 The concentration of carbon-14 in a living tree (remains about the same / keeps increasing / keeps decreasing). This concentration (remains about the same / starts to increase / starts to decrease) after the tree dies.

5 *A* is a sample of 1 mol of carbon-13 atoms and *B* is a sample of 1 mol of carbon-14 atoms.

(a) Which sample contains more carbon atoms?

(b) Which sample has a larger mass?

6 Complete the following table. Given: $N_A = 6.02 \times 10^{23}$ mol^{-1}

		Molar mass / g	Mass / g	Number of moles / mol	Number of atoms
(a)	C-14	14.0	0.2		
(b)	Th-238	238			2.35×10^{21}
(c)	I-127	127		0.05	

Exercise 2.3

★**1** Which of the following statements about the application of radionuclides is correct?

A A β source with a long half-life is used in thickness gauges so that it is not required to be replaced often.

B The γ source used in radiotherapy does not damage healthy cells.

C The α source in smoke detectors has a weak ionizing power so it is safe to look at it closely.

D An α source can be used to kill bacteria in food.

★**2** Ext A scientist finds that the activity of 20 g of carbon from an ancient animal bone is 0.5 Bq. Assume 1 g of carbon from a living animal has an activity of 0.267 Bq. Given that the half-life of carbon-14 is 5730 years, estimate the age of the bone.

A 5190 years

B 13 600 years

C 19 600 years

D 29 900 years

3 A cup of vegetable beef soup contains 750 mg of Na-23 atoms. The molar mass of Na-23 is 23.0 g. Given: $N_A = 6.02 \times 10^{23}$ mol^{-1}

(a) Estimate the number of moles of Na-23 in the soup.

(b) Hence, estimate the number of Na-23 atoms in it.

4 A sample contains 0.1 mol of carbon-13 atoms. The molar mass of carbon-13 13.0 g. Given: $N_A = 6.02 \times 10^{23}$ mol^{-1}

(a) Estimate the total mass of the carbon-13 atoms in the sample.

(b) Find the number of carbon-13 atoms in the sample.

5 An aluminium sheet is dragged between a radioactive source and a GM tube (Fig a).

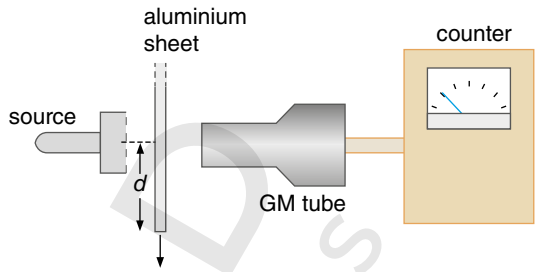


Fig a

The count rates recorded are shown in Table a. Let d be the distance between the source and the front end of the sheet.

d / cm	10	20	30	40	50	60	70	80
Count rate / s^{-1}	18	12	59	76	90	67	34	12

Table a

From the above data, what can you say about the thickness of the sheet between d = 30 cm and d = 60 cm?

★**6** In a study of plants' uptake of phosphorus, a liquid containing radioactive phosphorus-32 and stable phosphorus-31 is added to the soil. Some time later, a GM counter is used to inspect the plant growing in the soil (Fig b). A high count rate is detected when the GM tube is placed near the veins of the plant.

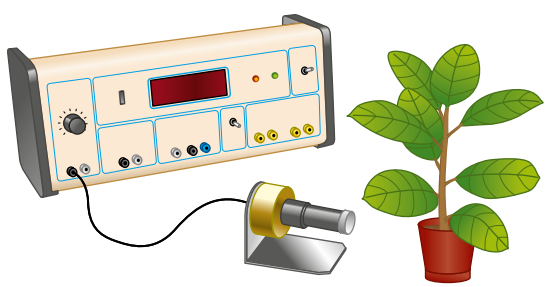


Fig b

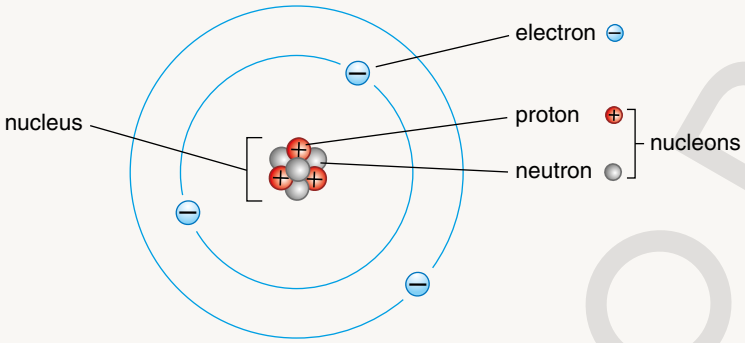
(a) What can you conclude from this observation?

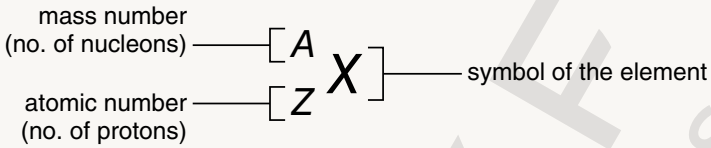
(b) Explain why α sources are not suitable for this kind of study.

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Key concepts 2

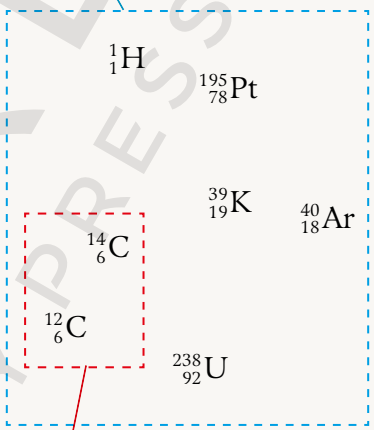
1 Atomic model





Nuclides

Atoms of particular mass numbers and atomic numbers



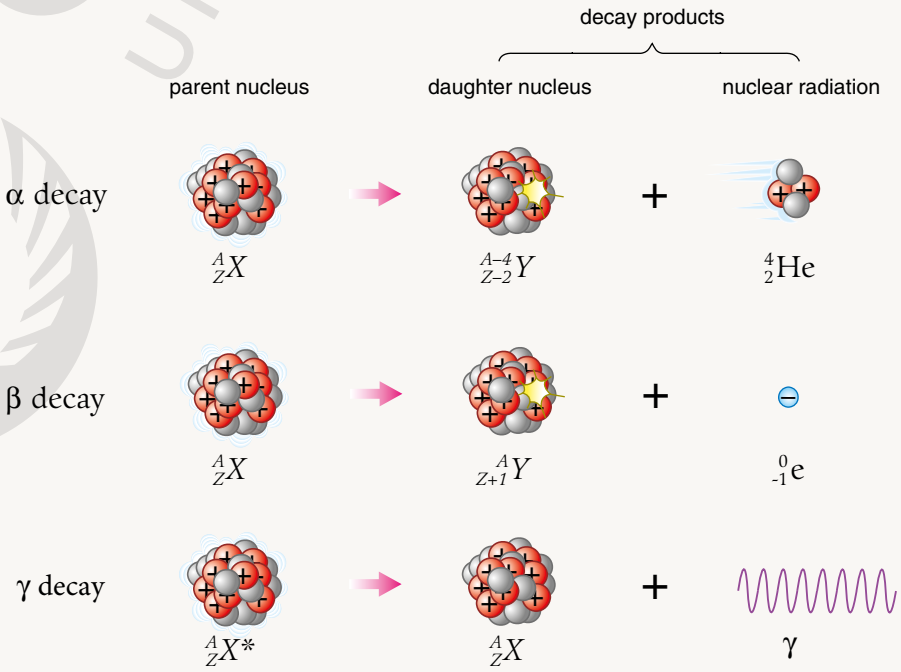
Isotopes

- Same atomic number
- Different mass numbers

2 Radioactive decay

Unstable nuclei emit α particles, β particles or γ rays to become more stable.

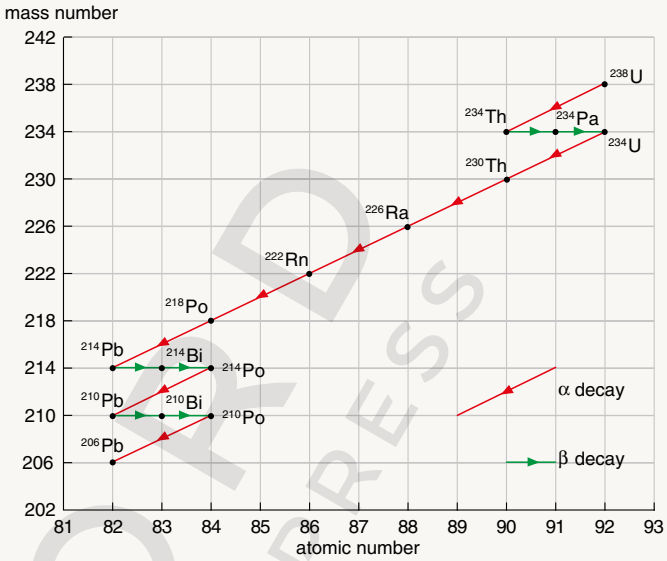
Radioactive decay is random in nature.



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3 Decay series

A nucleus may undergo a series of decays until a stable nucleus forms.



4 Decay curve

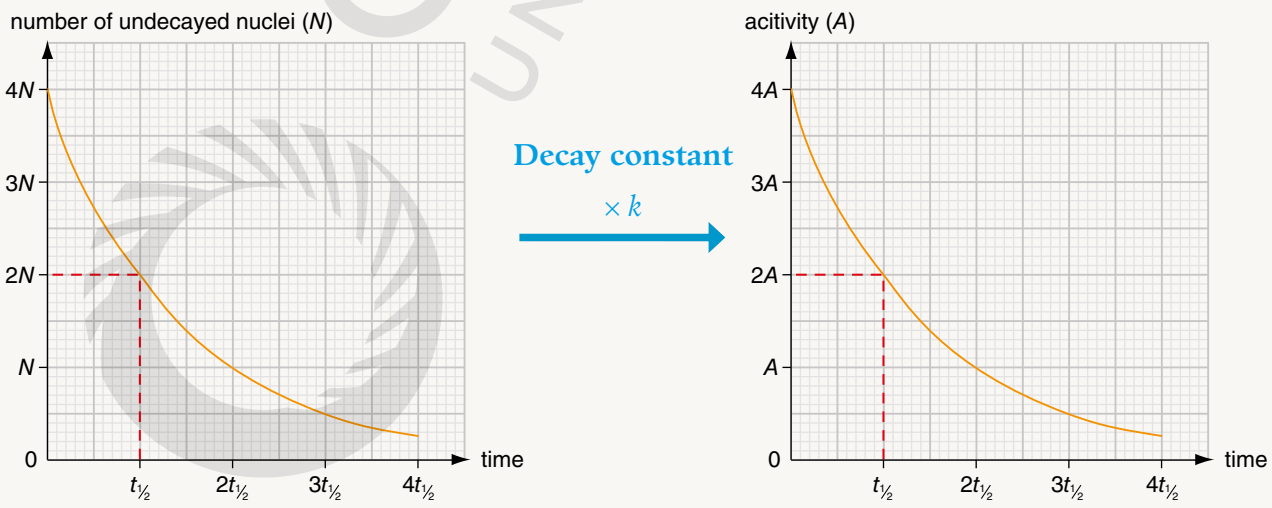
Related formulae

$$N = N_0\left(\frac{1}{2}\right)^n \qquad A = A_0\left(\frac{1}{2}\right)^n \qquad A = kN$$

Ext

$$N = N_0 e^{-kt} = N_0\left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}} \qquad A = A_0 e^{-kt} = A_0\left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}} \qquad t_{1/2} = \frac{\ln 2}{k}$$

Activity
= number of decays per unit time



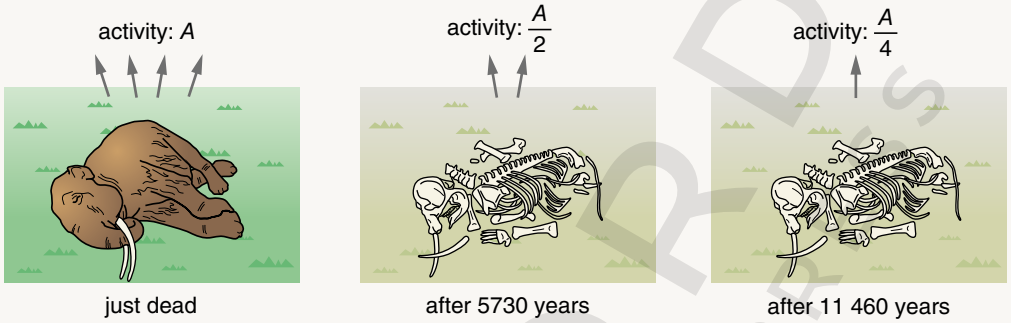
Half-life
= time taken for N to decrease by half
= time taken for A to decrease by half

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5 Uses of radionuclides

Examples of uses of radionuclides

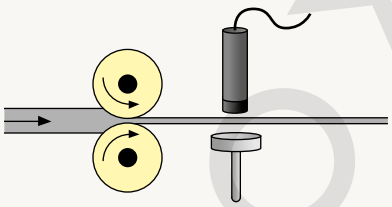
- Carbon-14 dating



- Radioactive tracer
- Killing cancer cells or microorganisms

Factors to consider when choosing radionuclides

- Ionizing power
- Penetrating power
- Half-life



Key terms

1	activity 放射強度	p.59
2	alpha decay (α decay) α衰變	p.50
3	atomic number 原子序數	p.47
4	becquerel (Bq) 貝克	p.59
5	beta decay (β decay) β衰變	p.50
6	carbon-14 dating 碳-14年代測定法	p.74
7	decay constant 衰變常數	p.59
8	decay curve 衰變曲線	p.60
9	electron 電子	p.44
10	gamma emission (γ emission) γ發射	p.50
11	half-life 半衰期	p.59
12	isotope 同位素	p.48
13	mass number 質量數	p.47
14	neutron 中子	p.44
15	nucleon 核子	p.45
16	nuclide 核素	p.48
17	proton 質子	p.44
18	radioactive decay 放射衰變	p.50
19	radionuclide 放射性核素	p.50
20	transmutation 嬗變	p.51

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Revision 2

Multiple-choice questions

1 2.1 Which of the following statements about the structure of an atom is correct?

A All the positive charges of an atom are concentrated in the nucleus.

B Electrons are held together by the attractive force they exert on each other.

C Both protons and neutrons are charged.

D The masses of protons and electrons are very small when compared to that of neutrons.

2 2.1 How many nucleons are there in the nucleus of ${}^{A}_{Z}X$?

A A

B Z

C $A - Z$

D $A + Z$

★**3** 2.2 Refer to the following graph (Fig a).

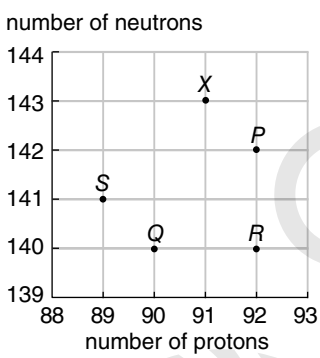


Fig a

If nuclide X undergoes an α decay followed by a β decay, which of the following will be formed?

A P

B Q

C R

D S

★**4** 2.3 Which of the following objects is suitable for using carbon-14 dating to estimate its age?

A Coal

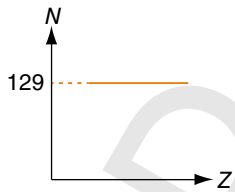
B Rock

C Ancient copper cooking utensil

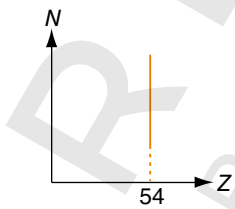
D Ancient ceramic vase

★**5** 2.1 N and Z represent the neutron number and the atomic number respectively. Which of the following graphs do isotopes of ${}^{129}_{54}Xe$ lie on?

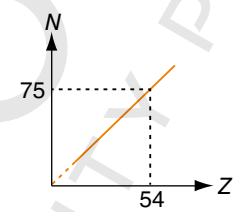
A



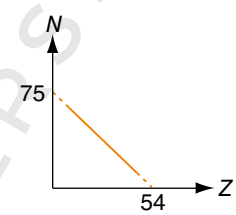
B



C



D



★**6** 2.2 Radionuclide X contains 83 protons and 130 neutrons. Which of the following are the possible daughter nuclei of X (Fig b)?

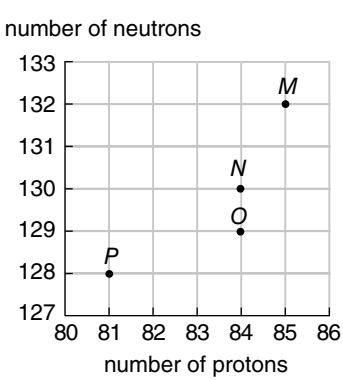


Fig b

A M and N

B M and O

C N and P

D O and P

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***7** 2.2 A decay series is shown below:

$$ {}^{i}_{P}X \xrightarrow{\beta} {}^{Q}_{R}Y \xrightarrow{\alpha} {}^{k}_{j}Z $$

What are the values of P, Q and R?

	P	***Q***	***R***
A	$j - 2$	i	$j + 2$
B	$j - 1$	i	$j + 1$
C	j	$k + 4$	$j - 1$
D	$j + 1$	$k + 4$	$j + 2$

***8** Ext 2.2 The activity of a radioactive sample drops by 20% in 2 hours. How long does it take for the activity to drop by 70%?

A 5 hours B 7 hours

C 9 hours D 11 hours

***9** Ext 2.2 The decay curves of two radioactive samples P and Q are shown below (Fig c).

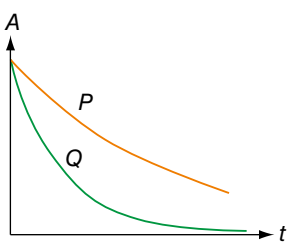


Fig c

Which of the following statements are correct?

(1) P has a longer half-life than Q.

(2) P has a larger decay constant than Q.

(3) A larger number of decays occur in sample Q than in sample P within one second.

A (1) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

***10** 2.2 Initially, a radioactive source contains M undecayed nuclei and its activity is A. After some time, N nuclei have decayed into a stable daughter nuclei and the activity of the source becomes a. The decay constant of the source is k. Which of the following equations is correct?

A $A = kN$ B $A = k(M - N)$

C $a = kN$ D $a = k(M - N)$

****11** 2.2 An initial count rate of R is recorded when a radioactive sample is placed near a GM counter. After one half-life, the count rate drops to 493 counts per minute. After another half-life, the count rate becomes 268 counts per minute. Suppose the background count rate is **not** negligible. Estimate the value of R (measured in counts per minute).

A 720

B 940

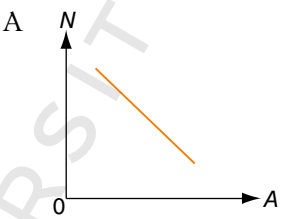
C 990

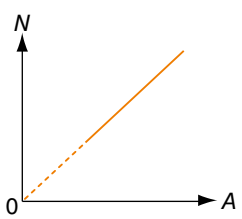
D 1070

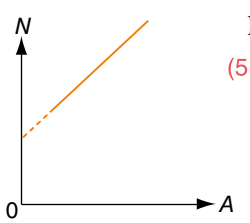
Refer p.61, 63

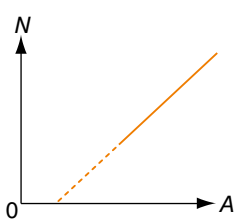
12 2.1 **HKDSE 2012 Paper 1A Q36**

Isotopes of an element have different mass number A and neutron number N. Which of the following N–A plots correctly shows the relationship of N and A for any given element?

A 

B 

C 

D (51%) 

13 2.2 **HKDSE 2018 Paper 1A Q32**

X and Y are two radioactive nuclides. The ratio of the mass of an atom of X to that of an atom of Y is 1 : 2. The half-lives of X and Y are T and $2T$ respectively. If two samples consisting of purely X and Y respectively have the same initial mass, find the ratio of the number of undecayed nuclei of X to that of Y after a period of $4T$.

A 1 : 4 (36%)

B 1 : 2 (40%)

C 1 : 1 (13%)

D 2 : 1 (11%)

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14 **HKDSE 2020 Paper 1A Q31**
2.2

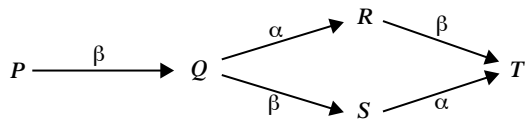


Fig d

Nuclide P can decay into nuclide T through either process $P - Q - R - T$ or process $P - Q - S - T$ as shown (Fig d). Which deductions below are correct?

(1) P and T are isotopes of the same element.

(2) Q and S have the same number of protons.

(3) S has one more neutron than R.

A (1) and (2) only

B (1) and (3) only (55%)

C (2) and (3) only

D (1), (2) and (3)

15 **HKDSE 2020 Paper 1A Q32**
2.2

The decay constant of a radioisotope of an element

A is random. (22%)

B depends on pressure and temperature. (8%)

C is directly proportional to the number of nucleons in the isotope. (21%)

D is an identifying characteristic of that isotope. (49%)

16 **HKDSE 2020 Paper 1A Q33**
2.2

Two radioactive samples P and Q are freshly prepared. It is found that when $\frac{15}{16}$ of all the nuclei of P have decayed, $\frac{63}{64}$ of all the nuclei of Q have also decayed. Find the ratio $\frac{\text{half-life of } P}{\text{half-life of } Q}$.

A 1 : 4

B 2 : 3

C 3 : 2 (38%)

D 4 : 1

Conventional questions

***17** 2.2 Astatine-219 ($^{219}_{85}$At) decays to form bismuth-215 ($^{215}_{83}$Bi).

(a) What kind of decay does $^{219}_{85}$At undergo? (1 mark)

(b) $^{215}_{83}$Bi is unstable and undergoes further decays. Two α particles and three β particles are emitted until a stable nuclide is formed. Find the atomic number and the mass number of the stable nuclide. (2 marks)

***18** 2.3 Some smoke detectors contain two metal plates and an α source (Fig e). Under normal conditions, the α particles emitted ionize the air molecules between the metal plates and allow a current to flow between them. When smoke particles enter the detector, the current drops. This triggers the alarm to sound.

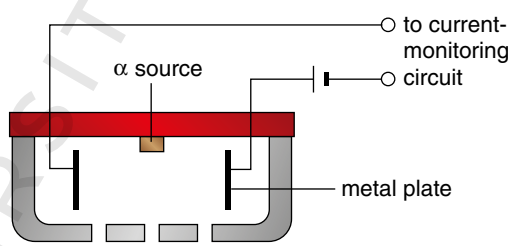


Fig e

(a) Explain why the current drops when smoke particles enter the detector. (2 marks)

(b) Give **two** reasons to explain why a γ source is not used in smoke detectors. (2 marks)

(c) Is it suitable to use a radioactive source with a short half-life? Explain . (2 marks)

***19** 2.3 Archaeologists sometimes use carbon-14 dating to find out the age of ancient remains. Carbon-14 has a half-life of 5730 years.

(a) Explain why the concentration of carbon-14 in living organisms is about constant. (2 marks)

(b) Would the method be useful for dating your physics textbook? Explain . (2 marks)

Ext (c) The activity of 20 g of carbon in a piece of wood is 7 disintegrations per minute. It is known that the activity of 1 g of carbon in a living plant is 16 disintegrations per minute. Estimate the age of the wood. (3 marks)

★**20** 2.2 Radioactive nuclide *A* undergoes a number of decays to form a stable nuclide *D* (Fig f). The half-life of *A* is 14 billion years.

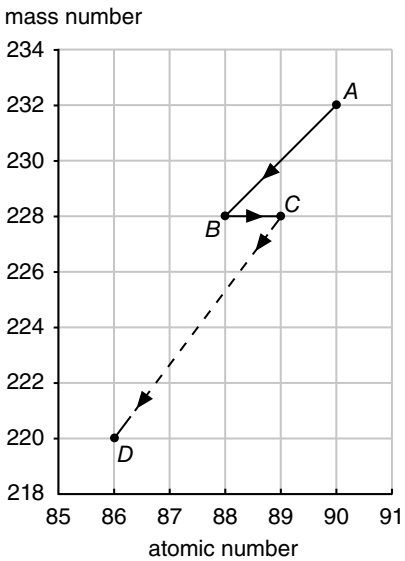


Fig f

(a) How many neutrons does *A* have? (1 mark)

(b) How many α particles are emitted during the part of the decay series from *C* to *D*? (1 mark)

(c) Are *B* and *C* isotopes of the same element? Explain your answer. (2 marks)

(d) A sample of *A* is placed in front of a GM tube. The activity of the sample is measured three times (Table a).

	Count rate / counts per minute
1st record	1217
2nd record	1203
3rd record	1198

Table a

A student claims that the count rate drops mainly because the number of nuclide *A* decreases. Do you agree? Explain your answer. (3 marks)

★**21** Misc In a dairy factory, a radioactive source is used to monitor the volume of milk in a box (Fig g). Normally, a box of milk should be filled to a certain level. When it passes between the source and the detector, the detector is just below the level of the milk.

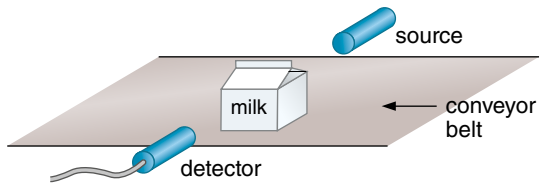


Fig g

(a) How would this set-up identify the boxes with insufficient volume of milk? (2 marks)

(b) Two radioactive sources, *X* and *Y*, are available for this purpose. Figure h shows how their activities vary with time.

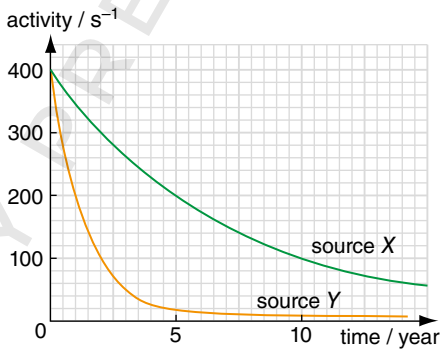


Fig h

(i) Find the half-lives of *X* and *Y*. (2 marks)

(ii) Which source has a half-life more suitable for monitoring the volume of milk in the above set-up? Explain briefly. (2 marks)

★**22** 2.3 In radiotherapy treatment, a radioactive source emitting γ rays is made to rotate around a patient's body to kill cancer cells (Fig i).

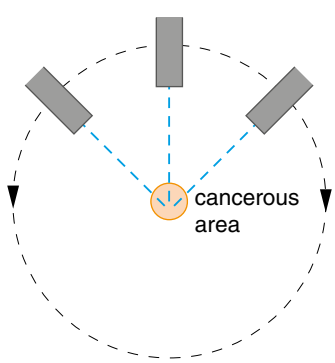


Fig i

(a) Explain why α radiation and β radiation are not suitable for this therapy. (1 mark)

(b) What will happen if the radioactive source is fixed in position instead of rotating around the patient's body? (2 marks)

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★23 2.3 A radioactive source is used in a factory to monitor the thickness of aluminium sheets produced (Fig j). The standard thickness should be 2 mm.

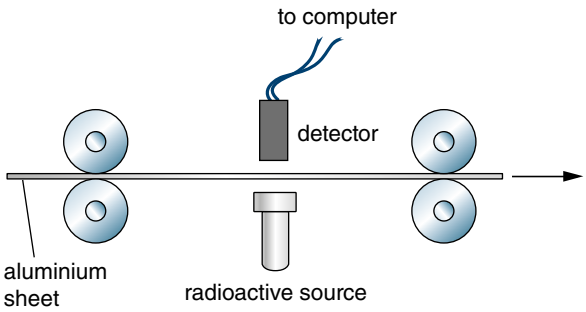


Fig j

(a) Suggest which type of radioactive source (α, β or γ) should be used. Explain briefly. (4 marks)

(b) When an aluminium sheet of standard thickness passes the detector, the count rate recorded is about 60 counts per second. State how the count rate changes when the aluminium sheet is thicker than 2 mm. (1 mark)

(c) A worker at the factory finds that even if the thickness of the aluminium sheet is exactly 2 mm, the count rate detected slightly fluctuates. Explain briefly. (1 mark)

★24 Ext Misc The age of some rocks can be determined by lutetium-hafnium dating. Lutetium-176 (Lu-176) undergoes β decay to form hafnium-176 (Hf-176) with a decay constant of $k = 5.92 \times 10^{-19}$ s^{-1}. Suppose that in a rock sample, the initial numbers of Lu-176, Hf-176 and Hf-177 atoms are x_0, y_0 and z_0 respectively. After time t, the numbers become x, y and z respectively. Hf-176 and Hf-177 are both stable.

Given: 1 year = 3.16×10^7 s

(a) The atomic number of lutetium is 71. Write an equation to represent the decay of Lu-176. (1 mark)

(b) (i) Express x in terms of x_0, k and t. (1 mark)

(ii) Express y in terms of y_0, x, k and t. (2 marks)

(iii) Hence, show that $\frac{y}{z} = \frac{y_0}{z_0} + \frac{x}{z}(e^{kt} - 1)$. (1 mark)

(c) It is determined through chemical methods that $\frac{y}{z} = 0.287$, $\frac{x}{z} = 0.04$ and $\frac{y_0}{z_0} = 0.284$. Estimate the age of the rock (in years). (2 marks)

★25 Misc Read the following passage about **radon gas** and answer the questions that follow.

Radon is a naturally occurring radioactive gas that you cannot see, smell or taste. It is formed when radium (Ra) in soil, rocks, water and building materials undergoes α decay. It may enter your home through cracks and joints in the building. It will be trapped and accumulate inside your home if the ventilation is poor.

One isotope radon-222 ($^{222}_{86}$Rn) decays and forms radioactive polonium (Po) (atomic number 84). If radon-222 is inhaled, its decay product will stay inside the body and undergo further radioactive decay until stable lead is formed. Long-term exposure to radon gas poses a higher risk of cancer.

(a) Write an equation for the decay from radium to $^{222}_{86}$Rn. (1 mark)

(b) Suggest a way how radon gas gets into your home. (1 mark)

(c) Why is radon-222 a health risk when it is inhaled into the body? (2 marks)

(d) Suggest a method to reduce the indoor radon level in your home. (1 mark)

(e) Suggest the most common type of cancer that can be caused by radon. (1 mark)

★26 Misc The thyroid is an organ located in the neck. Iodine-131 ($^{131}_{53}$I) can be used for a thyroid scan (Fig k). It is radioactive and decays to form stable xenon ($^{131}_{54}$Xe).



Fig k

(a) (i) What kind of decay does iodine-131 undergo? (1 mark)

(ii) Write down an equation for the decay. (1 mark)

Ext (b) In a thyroid scan, a sample of iodine-131 with an activity of 7.00×10^7 Bq is injected into a patient's body. Its activity drops to 6.85×10^7 Bq after 6 hours.

(i) Find the decay constant of iodine-131. (2 marks)

(ii) Find the half-life of iodine-131. (2 marks)

(c) The patient should stay at home for a few days after the scan. Suggest one reason why he should do this. (1 mark)

(d) Name two other sources of nuclear radiation in everyday life. (2 marks)

★★27 Misc Radioactive sodium-24 (Na-24) can be used to estimate the volume of blood in a human body. It is a β source with a half-life of 15 hours. A doctor injects some Na-24 of activity 2×10^4 Bq into the bloodstream of a patient (Fig l). After 30 hours, a 5 cm^3 sample of blood is taken from the patient.



Fig l

(a) (i) Estimate the activity of the Na-24 after 30 hours. (2 marks)

(ii) The activity of the blood sample is measured to be 6 Bq. Estimate the total volume of blood in the patient's body. (2 marks)

(iii) State one source of error in this estimation. (1 mark)

(iv) Explain why the blood sample is not taken immediately after the injection of Na-24. (1 mark)

(b) (i) Na-22 and Na-25 are also β sources with half-lives of 2.6 years and 59 s respectively. Explain why they are not suitable for the above purpose. (2 marks)

(ii) Explain how ionizing radiation can harm living cells in the human body. (2 marks)

▶ Refer p.63

★★28 Misc (a) Parts of the decay series of uranium-235 (U-235), uranium-238 (U-238) and thorium-232 (Th-232) are shown below.

U-235: … → $^{215}_{84}$Po → $^{211}_{82}$Pb → $^{211}_{83}$Bi → … → *A*

U-238: … → $^{218}_{84}$Po → $^{214}_{82}$Pb → $^{214}_{83}$Bi → … → *B*

Th-232: … → $^{216}_{84}$Po → $^{212}_{82}$Pb → $^{212}_{83}$Bi → … → *C*

A, *B* and *C* represent the end products in the series.

(i) During the part of the decay series from $^{235}_{92}$U to $^{211}_{83}$Bi, how many α particle(s) and β particle(s) are emitted? (2 marks)

(ii) The three stable end products are actually $^{206}_{82}$Pb, $^{207}_{82}$Pb and $^{208}_{82}$Pb. Match them with *A*, *B* and *C*. (2 marks)

Ext (b) In 1909, Rutherford and Royds showed that α particles are actually helium ions. Figure m shows schematically a part of their experimental set-up. A small closed cylinder *P* was fixed at the centre of a larger closed cylinder *Q*. All the air was removed from *Q*.

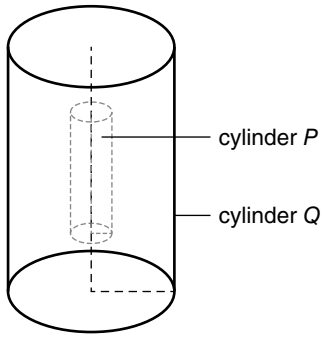


Fig m

At the beginning, a pure radium-226 (Ra-226) was put in *P*. The α particles emitted by Ra-226 could pass through the wall of *P* and entered *Q*. They gained electrons in the process and became neutral atoms. The experiment lasted for 5 days. Further investigation showed that helium gas existed in *Q*. This showed that α particles, after losing their charges, are helium atoms.

Suppose there are 6.20×10^{-4} mol of Ra-226 in *P* at the beginning. The decay constant of Ra-226 is 1.37×10^{-11} s^{-1}.

(i) The wall of *P* is made of glass. Explain why the glass wall should be very thin. (1 mark)

(ii) Find the half-life of Ra-226. Hence, explain why the activity of the Ra-226 is almost constant during the experiment. (3 marks)

(iii) Show that about 2.2×10^{15} α particles are emitted by the Ra-226 in 5 days. (3 marks)

Ext (c) Radium is used in medicine to produce radon gas, which is used for cancer treatment. Radon-222 (Rn-222) has a half-life of 3.825 days. It is prepared 24 hours prior to being administered to a patient. The activity is 6.00×10^{6} Bq when the patient receives the dose.

(i) Show that about 3.4×10^{12} Rn-222 nuclei are prepared. (3 marks)

(ii) Estimate the mass of Rn-222 needed to be prepared. (2 marks)

Refer p.67

29 HKDSE 2013 Paper 1B Q9

Misc

Carbon-14 dating can be used to identify the age of some objects which have the ^{14}C isotope, as it is radioactive and decays by emitting a β particle. A piece of wood sample is examined using carbon-14 dating and its activity is 0.2 Bq. The half-life of ^{14}C is 5730 years. Given: 1 year = 3.16×10^{7} s

Ext (a) Calculate the decay constant of ^{14}C in s^{-1}. Hence find the number of ^{14}C nuclei in this wood sample. (3 marks)

Assume that living organisms contain a constant proportion of carbon-14 in the ratio of $^{14}C/^{12}C = 1.3 \times 10^{-12}$ during its life time via intake of carbon dioxide (CO_2) from the atmosphere.

(b) The carbon content of this wood sample is found to contain a total of 1×10^{23} carbon nuclei. Estimate the number of ^{14}C nuclei in the sample originally when it died. (1 mark)

Ext (c) Estimate the age of this wood sample in years using the results found in (a) and (b). (2 marks)

See p.90 for the Exam Report.

30 HKDSE 2016 Paper 1B Q9

Misc

Part of the decay series of uranium-238 (U-238) is shown below. The end product lead-206 (Pb-206) is stable.

$$^{238}_{92}U \xrightarrow[4.5 \times 10^{9} \text{ years}]{\alpha} Th \xrightarrow[21 \text{ days}]{\beta} Pa \rightarrow \ldots \rightarrow {}^{206}_{82}Pb$$

(a) When a U-238 nucleus decays to a Pb-206 nucleus, how many α particle(s) and β particle(s) are emitted? (2 marks)

(b) As the first decay in the above chain from U to Th has a half-life much longer than those of subsequent decays, the decay from U-238 to Pb-206 can be simplified to **a single decay** with half-life 4.5×10^{9} years:

$$^{238}_{92}U \xrightarrow[4.5 \times 10^{9} \text{ years}]{} {}^{206}_{82}Pb$$

Suppose that a uranium-bearing rock contains only U-238 and no Pb-206 at the time when it was formed long ago by solidification of molten material. In a particular sample of the rock, it is found that that ratio $\frac{\text{number of Pb-206 atoms}}{\text{number of U-238 atoms}}$ is $\frac{2}{3}$ **at present**.

(i) Estimate the age of the rock. Assume that all Pb-206 atoms come from the decay of U-238 originally present in the sample and ignore the small number of U-238 atoms which have decayed but have not yet become Pb-206. (2 marks)

(ii) State, with a reason, whether the answer in (b)(i) is an overestimate or an underestimate of the age of the rock if some Pb-206 atoms have actually been lost. (2 marks)

Q29 Exam Report: In (a), candidates knew how to find the decay constant and the number of C-14 nuclei by different methods. In (c), some candidates confused *N* and N_0 or *A* and A_0, thus wrongly substituted the number of C-14 nuclei into the formula and failing to find the correct age of the wood sample.

(iii) The graph in Figure n shows how the number of U-238 atoms in the sample varies with time t subsequently while $t = 0$ denotes the **present time**. On Figure n, sketch a graph to show the variation of the number of Pb-206 atoms in the sample with time. (2 marks)

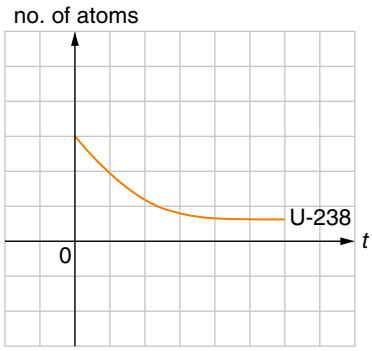


Fig n

See p.90 for the Exam Report.

31 HKDSE 2017 Paper 1B Q10

Misc

Dust may adhere to the surfaces of photos and films due to electrostatic attraction. To remove the dust effectively, a special brush with a thin slice of polonium-210 ($^{210}_{84}$Po) fixed near the brush hair as shown in Figure o may be used. Polonium-210 undergoes α decay and the daughter nucleus lead (Pb) is stable.

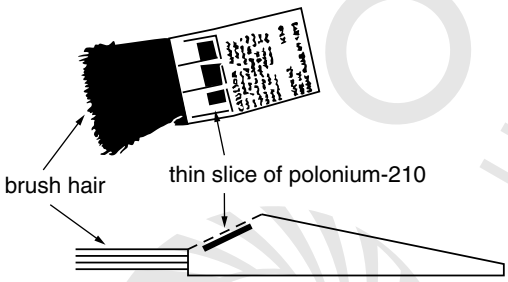


Fig o

(a) Write a nuclear equation for the decay of polonium-210. (2 marks)

(b) Briefly explain how the α particles help clean the charged dust. (2 marks)

(c) Briefly explain why the polonium-210 slice must be fixed near to the brush hair. (1 mark)

Ext (d) The manufacturer recommends that the brush should be returned to the factory for replacement of the polonium-210 slice every year. Taking the activity of a newly replaced polonium-210 slice as 1 unit, find its activity after one year (365 days). Given: half-life of polonium-210 is 138 days. (2 marks)

See p.90 for the Exam Report.

32 HKDSE 2018 Paper 1B Q10

Misc

(a) Part of the decay series of radium-226 (Ra-226) is shown below. Ra-226 decays to radon (Rn) by emitting an α particle with half-life 1600 years. The end product in the series is lead (Pb), which is stable.

$$^{226}_{88}\text{Ra} \xrightarrow{\alpha} \text{Rn} \longrightarrow \ldots \longrightarrow \text{Pb}$$

(i) $^{206}_{82}$Pb, $^{207}_{82}$Pb and $^{208}_{82}$Pb are three stable isotopes of lead. State, with a reason, which isotope can be the end product in this series. (2 marks)

Ext (ii) In a certain laboratory, a Ra-226 source has been used for 50 years. Estimate the percentage of undecayed Ra-226 left after this period. (2 marks)

(b) Spark counters can show the ionizing power of radiations. Figure p indicates the main features of a type of spark counters in school laboratories.

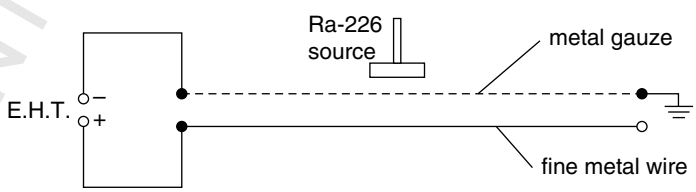


Fig p

A spark counter consists of a fine metal wire mounted a few mm beneath an earthed metal gauze. The wire is connected to the positive terminal of an E.H.T. (Extra High Tension) supply so that a very intense electric field is set up between the wire and the metal gauze. When a Ra-226 source is brought near the gauze, sparks giving out flashes of light and crackling sound are produced at irregular intervals.

(i) Explain why the sparks occur at **irregular** intervals. (1 mark)

A Ra-226 source used in school laboratories is usually said to emit α, β as well as γ radiations.

(ii) Explain why β radiation is also emitted even though the source is primarily an α emitter. (1 mark)

(iii) Why is the sparking mainly caused by α radiation rather than β or γ radiation? Suggest a simple way to verify this. (2 marks)

33 HKDSE 2021 Paper 1B Q9

Misc

Potassium-40 ($^{40}_{19}K$) is a natural radioisotope of potassium.

(a) (i) What kind of decay does $^{40}_{19}K$ undergo if it decays to $^{40}_{20}Ca$? (1 mark)

(ii) As banana is rich in potassium, a student claims that the radiation emitted by $^{40}_{19}K$ after eating a few bananas can be detected outside the human body. Explain whether this claim is justified. (1 mark)

Ext (b) A banana typically contains 0.45 g potassium in which 0.012% by mass is $^{40}_{19}K$ while the rest is $^{39}_{19}K$.

Given: half-life of $^{40}_{19}K = 1.25 \times 10^9$ years

1 year $= 3.16 \times 10^7$ seconds

molar mass of $^{40}_{19}K = 40.0$ g

(i) Estimate the number of moles of $^{40}_{19}K$ in a banana. (1 mark)

(ii) Deduce the activity, in Bq, of a banana. (2 marks)

Q33 Exam Report: This question tested candidates' knowledge and understanding of radioactivity in the context of radioactive potassium contained in bananas. The performance of candidates was good. In (a)(i), most candidates knew that β decay occurred. Candidates' performance was fair in (a)(ii) as some explained the answer in terms of ionizing power or range of β particles instead of penetrating power. Many candidates knew how to tackle part (b). However, quite a number of them made mistakes in unit conversions or mistook *N* as the number of moles of radionuclides.

Q30 Exam Report: This question tested candidates' understanding of radioactivity. Part (a) was well answered. Most candidates employed a correct formula in (b)(i) to find the age of the sample but not many were able to work out the correct ratio $\frac{N}{N_0} = \frac{3}{5}$ from the information given. In (b)(ii), the reasons stated by the candidates for the age found being an underestimate were mostly far from concise. In sketching the graph in (b)(iii), few candidates realized that the number of Pb-206 atoms is non-zero at $t = 0$.

Q31 Exam Report: This question on radioactivity was in general well answered. Candidates did well in parts (a) and (c). In (b), quite a number of candidates wrongly thought that the α particles neutralised the charged dust directly. Most candidates knew how to calculate the activity in part (d), only some of them made mistakes in the unit.

Q32 Exam Report: Part (a) was well answered though some candidates commented that atoms with more neutrons or greater mass numbers were more stable. In (b)(i), the answers of candidates were far from precise e.g. 'as the ionizing power is random or the type of radiation emitted is random'. Weaker candidates only focused on the EHT device. In (b)(ii), many candidates wrongly held that the Ra-226 source was not pure and thus emitted all three kinds of radiation. It seems that candidates did not fully understand the mechanism of a decay series. In (b)(iii), quite a number of them pointed out that the sparks were related to the strong ionizing power of α radiation, but many just suggested using a cloud chamber to distinguish different types of radiations emitted instead of describing a way to verify the sparks were caused by the α radiation.